

TICVAI

MARKETING & CRM

Backend Configuration Scope of Work

12 Boards | 120 Configuration Screens | High-Definition UI/UX Reference

Document type	Implementation Scope of Work and UI/UX Configuration Reference
Prepared for	TICVAI Product & Delivery Team / SoftLab Implementation Review
Version	1.0
Date	13 August 2026

AI-first. Multi-tenant. One connected platform.

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Document Purpose & Use

Purpose. This document translates the Marketing & CRM requirement matrix into twelve implementation boards. Each board contains one high-definition visual reference and a detailed Scope of Work for its ten backend configuration screens.

Design intent. The visuals communicate information architecture, hierarchy and expected configuration coverage. SoftLab shall develop production-ready responsive interfaces and may refine interaction details while preserving the approved business scope, shared architecture and TICVAI design system.

Delivery interpretation. A screen is not complete when only the visible mockup is reproduced. The delivered feature must include the described rules, validations, permissions, integrations, states, audit evidence and exception handling.

Platform-Wide Architectural Requirements

- One connected platform: Marketing & CRM shall reuse shared Customer Master, Consent, Promotion, Loyalty, Wallet, Ticketing, Reservation, Payment, Notification, Approval, Audit and AI services rather than duplicate them.
- Multi-tenancy: every applicable screen shall support tenant, brand, venue, region, legal entity, language, timezone and channel context, with strict data isolation.
- Security: access shall be controlled through SSO/MFA and RBAC/PBAC, including field masking, temporary/delegated access, segregation of duties and emergency override where approved.
- Governance: high-impact configuration changes shall support draft, validation, review, approval, publication, versioning, rollback and immutable audit history.
- AI governance: AI outputs shall be explainable, confidence-scored, reviewable and subordinate to consent, eligibility, capacity, pricing, finance and access-control policies.
- Integration: services shall expose secured, versioned APIs and events and implement idempotency, retry, monitoring, correlation IDs and error reconciliation.
- Privacy: consent, purpose limitation, minimization, retention, anonymization and data-subject rights shall be enforced across UI, API, exports, analytics and AI processing.
- Usability and accessibility: desktop-first administration shall use consistent TICVAI navigation and terminology, accessible contrast, keyboard support and WCAG 2.2 AA-aligned components.
- Operational quality: all grids shall support practical search, filter, sort, pagination, saved views and controlled export; all flows shall include empty, loading, warning, error and recovery states.
- Auditability: every configuration, approval, publication, AI recommendation, automated action, override, export and sensitive-data access shall record actor, timestamp, context, before/after value and reason where applicable.

Board Directory

Board	Configuration Area	Matrix Coverage
1	CRM Guest Master & Customer 360	section 22.2 - Guest Profiles & Data Management
2	Identity Resolution, Consent & Data Privacy	sections 22.2.8-22.2.9 and 22.13 - Consent Management
3	Audience Segmentation & Personalization	section 22.14 plus audience attributes in section 22.2
4	Campaign Management & Attribution	section 22.1 and attribution requirements in section 22.14
5	Marketing Automation & Customer Journeys	Journey-automation requirements following matrix section 22.3.10, plus event-triggered campaign requirements
6	Newsletter, Notifications & Message Delivery	sections 22.4 - Newsletter & Email Marketing and 22.9 - Notifications & Messaging
7	Omnichannel Inbox, AI Chatbot & Agent Workspace	section 22.8 - Omnichannel Communications & AI Chatbot
8	Case Management, SLA & Service Recovery	First ten requirements in matrix section 22.3 - Case Management & SLA
9	Surveys, Reviews & Voice of Customer	sections 22.5 - Reviews & Ratings and 22.7 - Surveys & Feedback
10	Gamification & Loyalty Engagement	section 22.6 - Gamification & Loyalty
11	CMS, White-Label Content & SEO	sections 22.10 - Marketing/CMS and 22.11 - SEO Management & Optimization
12	Digital Waivers & Signature Management	section 22.12 - Digital Waivers & Signature Management

Board 1 - CRM Guest Master & Customer 360

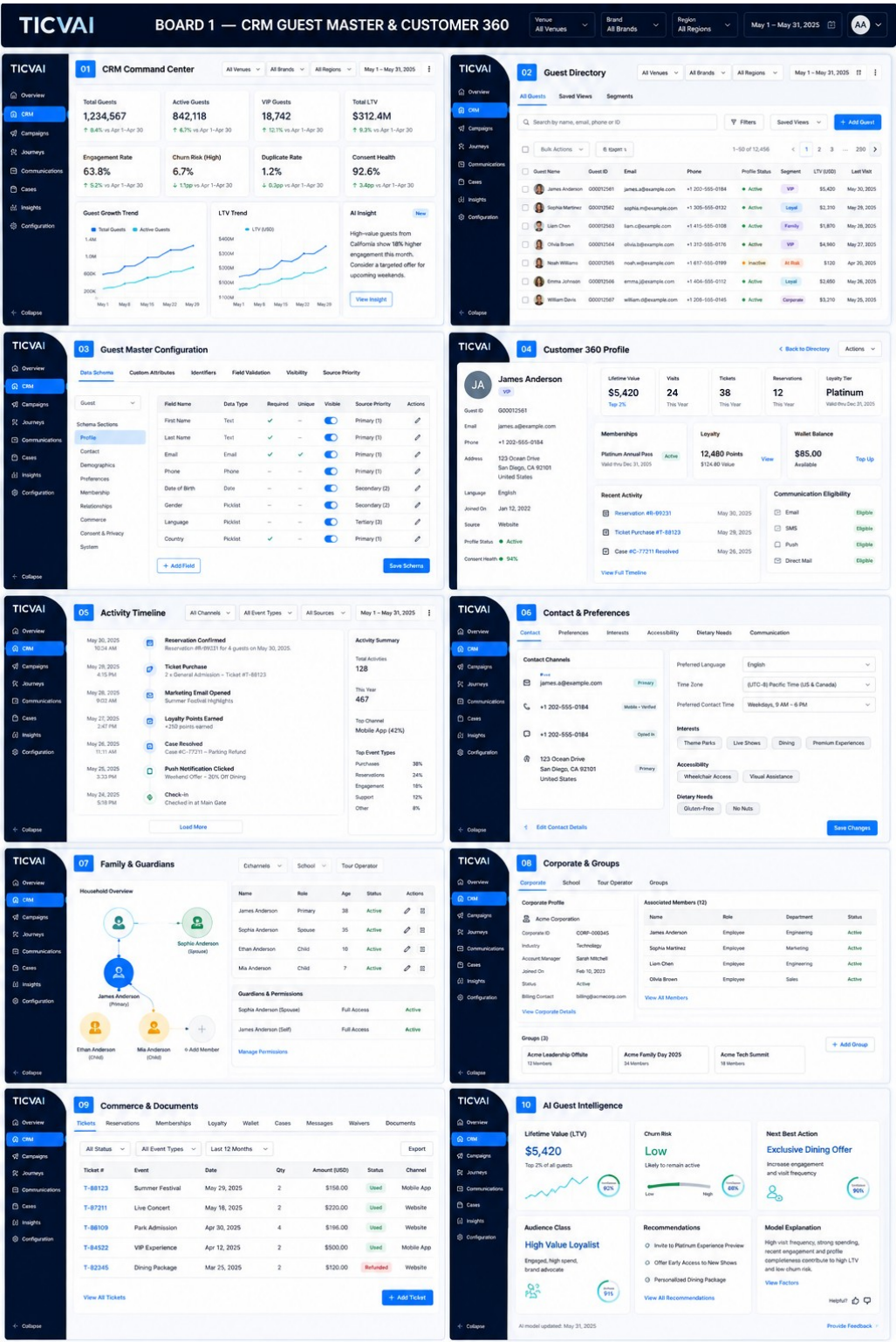


Figure 1. High-definition configuration board with all 10 screens.

Board 1 Scope of Work

Purpose. Establish the shared Customer Master Service and a single, trusted 360-degree guest record across every TICVAI sales, service, engagement and operational channel.

Source coverage. Matrix section 22.2 - Guest Profiles & Data Management

Operating workflow. Capture or receive guest data -> validate and resolve identity -> enrich the master record -> present the Customer 360 view -> calculate value, engagement and risk -> recommend the next best action.

Mandatory integrations. Ticketing, B2C, B2B, POS, mobile application, reservations, membership, loyalty, wallet, payments, F&B, retail, communications, cases, surveys, reviews, waivers, analytics and secured CRM APIs.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - CRM Command Center

Purpose: Give authorized users an operational and analytical overview of the customer base.

- Display total, new, active, inactive, registered and guest-checkout customers with tenant, brand, venue, region and date filters.
- Show VIP, high-value, family, corporate, group, churn-risk, duplicate and incomplete-profile indicators.
- Present LTV, engagement, growth, consent health and source-synchronization trends with drill-down to the underlying guests.
- Surface explainable AI insights and prioritized actions without bypassing consent, eligibility or access policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Guest Directory

Purpose: Provide a permission-controlled directory for locating and managing every guest record.

- Search by name, email, mobile, guest ID, external ID, ticket, booking, membership or loyalty identifier.
- Filter by profile status, segment, language, geography, membership, loyalty tier, wallet, LTV, engagement and churn risk.
- Support saved views, configurable columns, sorting, pagination, controlled export and auditable bulk actions.
- Open Customer 360 directly from a result while masking restricted personal or financial data by policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Guest Master Configuration

Purpose: Configure the shared guest data model without software changes.

- Maintain standard and custom attributes, field groups, labels, data types, defaults, required flags and validation rules.
- Define primary and external identifiers, source-system priority, survivorship rules and profile-completeness scoring.
- Configure field visibility and editability by role, tenant, brand, venue, region and jurisdiction.
- Version and audit schema changes and expose approved attributes consistently through UI, API and events.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Customer 360 Profile

Purpose: Present a consolidated, actionable view of one guest.

- Display identity, contacts, preferences, household, organization, membership, loyalty, wallet, tickets, bookings and visits.
- Show LTV, engagement, churn risk, communication eligibility, open cases, pending waivers and recent activity.
- Provide permission-controlled quick actions for communication, booking, case creation, campaign enrollment and profile maintenance.
- Allow role-based widget configuration while keeping the Customer Master Service as the system of record.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Activity Timeline

Purpose: Provide a chronological record of guest activity across the connected platform.

- Ingest purchases, ticket usage, reservations, visits, membership changes, loyalty, wallet, campaigns, messages, cases, surveys and waivers.
- Filter by date, channel, venue, event type, source system and outcome and open the related source transaction.
- Distinguish operational facts from user notes and AI-derived events and preserve event timestamps and source identifiers.
- Support reliable ordering, pagination and audit evidence without allowing historical events to be silently altered.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Contact & Preferences

Purpose: Maintain verified contact information and service preferences for each guest.

- Manage multiple email addresses, phone numbers, physical addresses, emergency contacts and social identifiers.
- Record preferred language, channel, contact time, frequency, interests, favorite attractions and visit preferences.
- Store accessibility and dietary requirements with suitable sensitivity and role restrictions.

- Show verification, suppression and consent status while delegating legal enforcement to the shared Consent Service.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Family & Guardians

Purpose: Model households, dependants and guardians while preserving each person's individual identity.

- Create family groups and link parent, guardian, spouse and dependant relationships with effective dates.
- Configure purchasing, booking, profile-management and waiver-signing authority for minors and dependants.
- Support shared or separate benefits, memberships, bookings and communications without merging individual profiles.
- Record additions, removals, authority changes and exceptions in a complete relationship audit trail.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Corporate & Groups

Purpose: Link guests to corporate, school, travel and group structures.

- Support corporate accounts, schools, tour operators, travel agencies, resellers, clubs, teams and event groups.
- Define contacts, participant roles, billing relationships, booking authority and relationship validity periods.
- Display associated bookings, memberships, agreements and activity while respecting organizational access boundaries.
- Expose relationships to B2B, reservations, cases and reporting through secured services and APIs.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Commerce & Documents

Purpose: Consolidate the guest's commercial, service and document history.

- Provide tabs for tickets, reservations, memberships, loyalty, wallet, refunds, exchanges, transfers, upgrades and attendance.
- Display communications, cases, survey responses, ratings, reviews, waivers, identification documents and signed agreements.
- Link every item to its source transaction and show status, value, channel, venue, timestamps and authorized actions.
- Apply RBAC/PBAC, masking, retention, download and audit policies to financial, identity and legal records.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - AI Guest Intelligence

Purpose: Turn the unified record into explainable customer insight and recommended actions.

- Calculate historical and predicted LTV, engagement, churn, inactivity, cancellation, affinity and upgrade propensity.
- Classify audiences and recommend the next campaign, offer, product, membership, reward, channel and next best action.
- Display contributing factors, confidence, model/version, expected impact and policy or data limitations.
- Require consent and eligibility checks, support accept/reject/feedback actions and audit all human and AI decisions.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 2 - Identity Resolution, Consent & Data Privacy

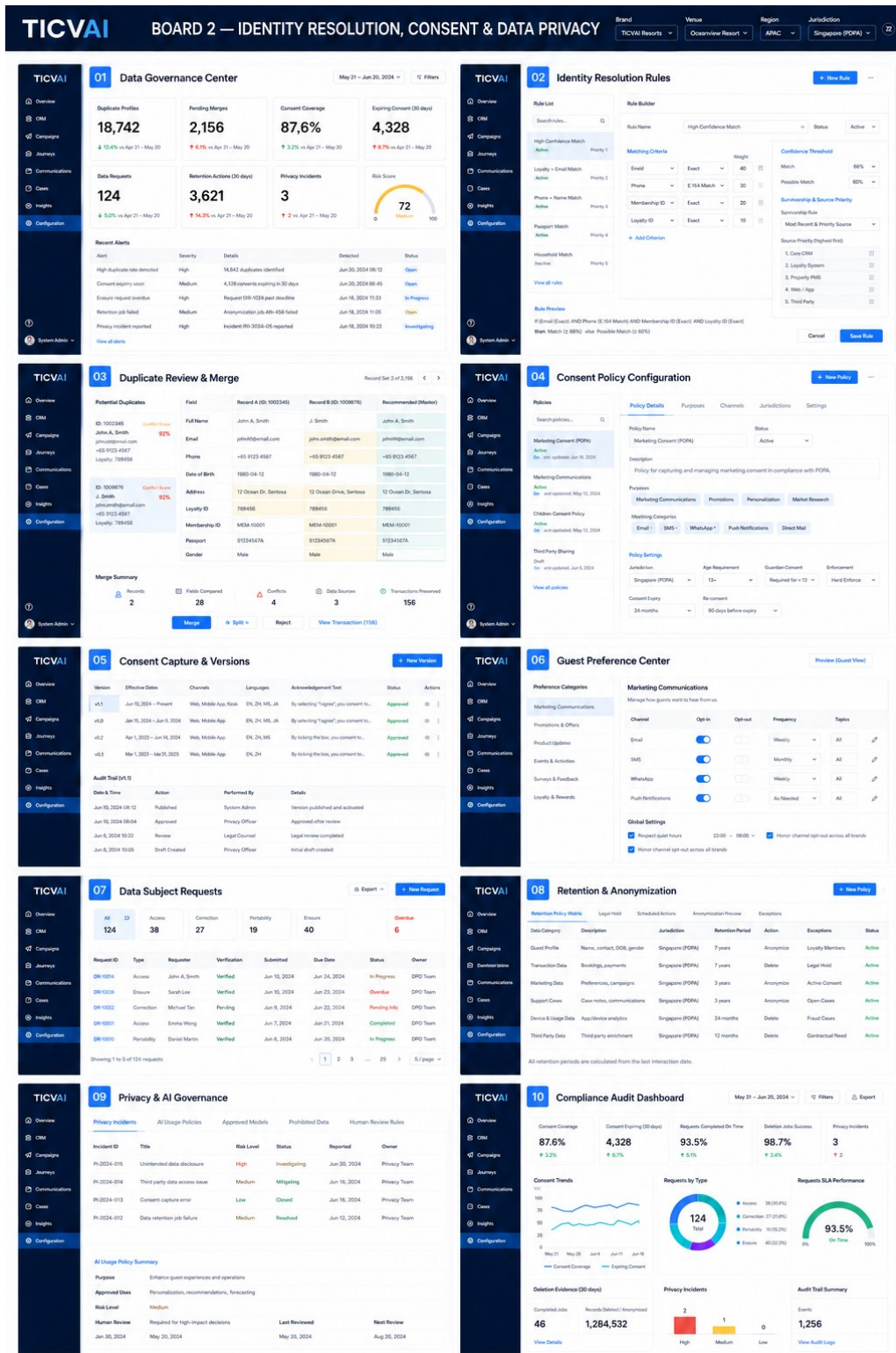


Figure 2. High-definition configuration board with all 10 screens.

Board 2 Scope of Work

Purpose. Provide controlled identity matching, consent capture and enforcement, privacy-rights workflows, retention and AI governance across the full TICVAI platform.

Source coverage. Matrix sections 22.2.8-22.2.9 and 22.13 - Consent Management

Operating workflow. Monitor data quality -> resolve possible duplicates -> capture versioned consent -> enforce preferences -> process privacy requests -> retain, anonymize or delete data -> demonstrate compliance.

Mandatory integrations. Customer Master, identity provider, B2C/B2B/POS/mobile/kiosk channels, campaigns, communications, analytics, document storage, security monitoring, case management and audit services.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Data Governance Center

Purpose: Provide a consolidated view of customer-data quality, consent and privacy operations.

- Show duplicate profiles, pending merges, consent coverage, expiring consent, data-subject requests, retention actions and incidents.
- Filter by tenant, brand, venue, region, jurisdiction, data category, severity and owner.
- Surface overdue work, legal deadlines and high-risk exceptions with drill-down to the governing record.
- Provide trend analysis and controlled exports for privacy, legal, security and audit teams.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Identity Resolution Rules

Purpose: Configure how potential duplicates are detected and a master identity is selected.

- Build weighted exact, normalized and fuzzy matching rules for email, phone, name, date of birth, membership, loyalty, passport and external IDs.
- Define match, possible-match and no-match thresholds, source priority, survivorship and field-level confidence.
- Support jurisdictional restrictions, excluded sources, false-positive handling and rule simulation before activation.
- Version, approve and audit rules and record the rule/version used for every automated decision.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Duplicate Review & Merge

Purpose: Allow trained users to review, merge, split or reject suspected duplicate profiles.

- Compare records side by side, highlight conflicts and display recommended master values with confidence and source provenance.

- Preview linked tickets, bookings, memberships, loyalty, wallet, cases, consents and documents before committing.
- Preserve all transaction history, identifiers, relationships and audit evidence and prevent unsafe automatic merges.
- Require reason codes and approval for high-risk merges and support controlled split/recovery where permitted.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 04 - Consent Policy Configuration

Purpose: Define the legal and operational rules governing consent.

- Configure consent purposes, marketing categories, processing purposes and channel-specific permission for email, SMS, WhatsApp, push and direct mail.
- Define jurisdiction, lawful basis, age and guardian requirements, capture channels, expiry, renewal and hard/soft enforcement.
- Map policies to brands, venues, products, audiences and processing activities with precedence and conflict handling.
- Require legal review, effective dates, version control and immutable policy-change history.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 05 - Consent Capture & Versions

Purpose: Manage the content and lifecycle of guest consent statements.

- Create multilingual consent versions with acknowledgement text, links, effective dates and supported channels.
- Capture who consented, what they saw, when, where, by which channel and under which policy version.
- Support withdrawal, expiry, re-consent and guardian authorization without overwriting prior evidence.
- Publish only approved versions and make the active statement available consistently to all touchpoints.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 06 - Guest Preference Center

Purpose: Configure the guest-facing center for communication choices and privacy preferences.

- Expose subscription categories, topics, preferred channels, contact frequency, quiet hours and global or brand-level opt-outs.
- Apply identity verification and clearly distinguish transactional messages from optional marketing.
- Synchronize changes in real time to campaigns, journeys, chatbot, newsletters and notification delivery.
- Store an auditable receipt for every preference change and provide accessible, multilingual presentation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Data Subject Requests

Purpose: Manage privacy requests from intake through verified completion.

- Support access, rectification, portability, erasure, restriction and objection request types.
- Verify requester identity, calculate statutory deadlines, assign owners, gather approvals and track dependencies.
- Search connected systems, assemble export packages and document exclusions, legal holds or denied actions.
- Record communications, evidence, completion status and SLA performance in an immutable case history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Retention & Anonymization

Purpose: Configure how long customer data is retained and what happens at expiry.

- Define policies by data category, jurisdiction, purpose, guest status, transaction type and source system.
- Support archive, anonymize, pseudonymize and delete actions, legal holds, fraud exceptions and scheduled execution.
- Preview affected records, dependencies and downstream consequences before high-impact actions.
- Produce completion evidence and reconcile results across analytics, search, documents, backups and integrations.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Privacy & AI Governance

Purpose: Control privacy incidents and approved uses of customer data by AI.

- Track incident type, severity, affected data, containment, owner, notification obligations, mitigation and closure.
- Maintain approved models/use cases, prohibited data, risk level, purpose limitation and human-review requirements.
- Configure prompt/output retention, masking, vendor/model access and escalation for sensitive or high-impact decisions.
- Link controls to AI recommendations across CRM and preserve model, policy, reviewer and outcome evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Compliance Audit Dashboard

Purpose: Demonstrate consent and privacy compliance through measurable evidence.

- Report consent coverage and expiry, request volumes and SLA, retention completion, anonymization, deletion and incident trends.
- Drill down by jurisdiction, business unit, channel, policy, owner and data category.
- Provide controlled evidence packages, exception logs and audit exports with source references and timestamps.
- Restrict compliance information appropriately and record every view, export and administrative action.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 3 - Audience Segmentation & Personalization

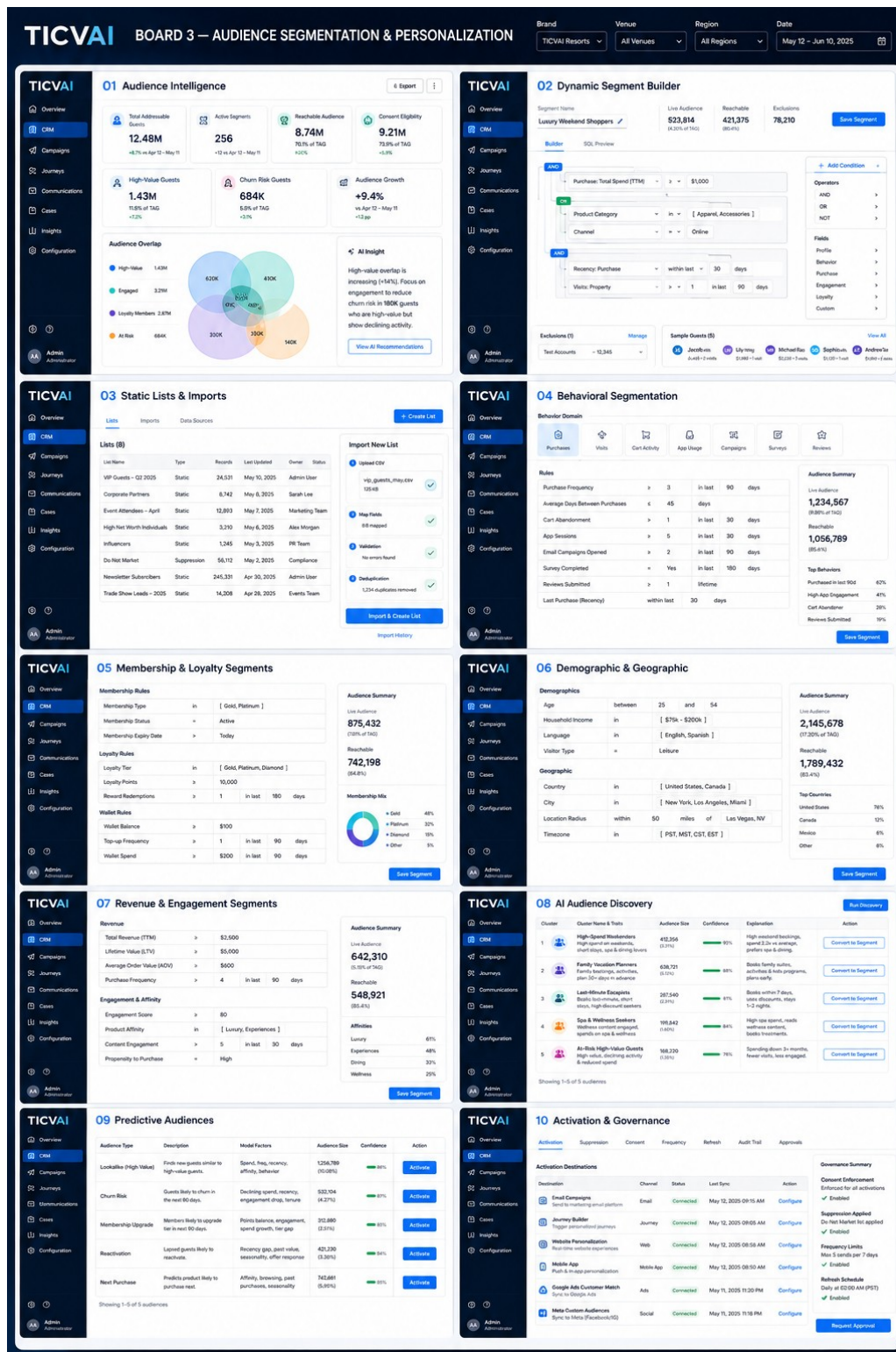


Figure 3. High-definition configuration board with all 10 screens.

Board 3 Scope of Work

Purpose. Create governed, reusable audiences from unified guest data and activate them safely across TICVAI marketing, content and service channels.

Source coverage. Matrix section 22.14 plus audience attributes in section 22.2

Operating workflow. Select attributes and behaviors -> build or discover an audience -> preview size and eligibility -> approve -> activate -> refresh -> monitor quality, overlap and attribution.

Mandatory integrations. Customer Master, consent, ticketing, membership, loyalty, wallet, reservations, commerce, campaigns, journeys, CMS, mobile app, analytics, CDP/advertising connectors and audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Audience Intelligence

Purpose: Give marketers and CRM teams an overview of addressable audience health.

- Show total guests, reachable guests, active segments, consent-eligible audience, high-value and churn-risk populations.
- Visualize growth, overlap, duplication, suppression and reachability by channel, brand, venue and region.
- Identify stale, shrinking, high-performing or conflicting audiences and link to the responsible definitions.
- Provide explainable AI observations without modifying or activating segments automatically.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Dynamic Segment Builder

Purpose: Allow authorized users to create automatically refreshed rule-based audiences.

- Provide nested AND/OR/NOT logic across profile, behavior, transaction, membership, loyalty, wallet and custom attributes.
- Show live audience count, reachable count, exclusions, consent impact and representative sample guests while rules are edited.
- Support effective dates, refresh frequency, ownership, tags, descriptions, approvals and reusable rule groups.
- Validate incompatible conditions, excessive complexity and restricted attributes before save or activation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Static Lists & Imports

Purpose: Manage governed fixed audiences and externally supplied lists.

- Create lists manually or through secure file import with field mapping, validation, deduplication and error handling.

- Match imported members to the Customer Master and control whether unmatched records may create leads or remain quarantined.
- Record source, owner, purpose, consent basis, expiry, refresh history and file provenance.
- Support additions, removals, suppression and export under RBAC/PBAC and audit controls.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Behavioral Segmentation

Purpose: Build audiences from observable guest activity.

- Use purchase, visit, ticket scan, reservation, cart, app, web, campaign, chatbot, survey and review events.
- Support recency, frequency, count, value, sequence, absence-of-event and time-window conditions.
- Preview top behaviors, sources and data freshness and exclude bot, employee, test or fraudulent activity.
- Refresh segments through event streams and expose rule/version provenance for each inclusion.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Membership & Loyalty Segments

Purpose: Create audiences using membership, loyalty and wallet state.

- Filter by membership type, status, start, expiry, renewal, freeze, benefit use and upgrade eligibility.
- Filter by loyalty tier, points, earn/redemption history, reward use, milestone and tier movement.
- Use wallet balance, top-up, spend, low-balance and dormancy conditions while protecting financial data.
- Combine these signals with profile and behavior rules and show reachable audience before activation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Demographic & Geographic

Purpose: Configure demographic and location-based audiences responsibly.

- Use age band, household, language, country, city, postcode, visitor type, nationality where lawful and custom demographics.
- Support location radius, venue proximity, travel market, timezone and resident/tourist classification.
- Display consent, fairness, minimum-audience and restricted-attribute warnings before use.
- Allow jurisdiction-specific field availability and prevent prohibited sensitive targeting.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Revenue & Engagement Segments

Purpose: Create value- and engagement-based audiences.

- Use total revenue, LTV, average order value, purchase frequency, category spend, refund behavior and propensity.
- Use engagement score, campaign interaction, app activity, visit frequency, inactivity and product affinity.
- Define bands, percentiles, scoring periods and calculation sources and preview distribution before publishing.
- Protect against circular attribution and record the data snapshot and scoring version used.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - AI Audience Discovery

Purpose: Discover meaningful customer clusters using explainable AI.

- Present suggested clusters with dominant traits, size, reachability, value, engagement and observed opportunity.
- Show contributing variables, confidence, stability, bias/fairness checks and excluded sensitive attributes.
- Allow users to inspect sample profiles and convert a recommendation into an editable governed segment.
- Require human approval and retain feedback, model/version and final segment definition.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Predictive Audiences

Purpose: Create model-based audiences for future behavior and opportunity.

- Support lookalike, churn, reactivation, membership upgrade, next purchase and high-value propensity audiences.
- Display prediction window, threshold, audience size, confidence, lift and primary model factors.
- Allow threshold simulation and exclusions and monitor drift, performance and fairness after activation.
- Never use predictive status to override consent, eligibility, pricing, capacity or service policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Activation & Governance

Purpose: Control where audiences are used and how they remain compliant.

- Activate to campaigns, journeys, website, mobile app and approved external platforms through secured connectors.
- Configure suppression, consent enforcement, frequency limits, destination mapping, refresh schedule and expiry.
- Require approval for sensitive, high-volume or external activation and provide a pre-flight impact summary.

- Track synchronization, failures, member counts, destination use, versions and the complete activation audit trail.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 4 - Campaign Management & Attribution

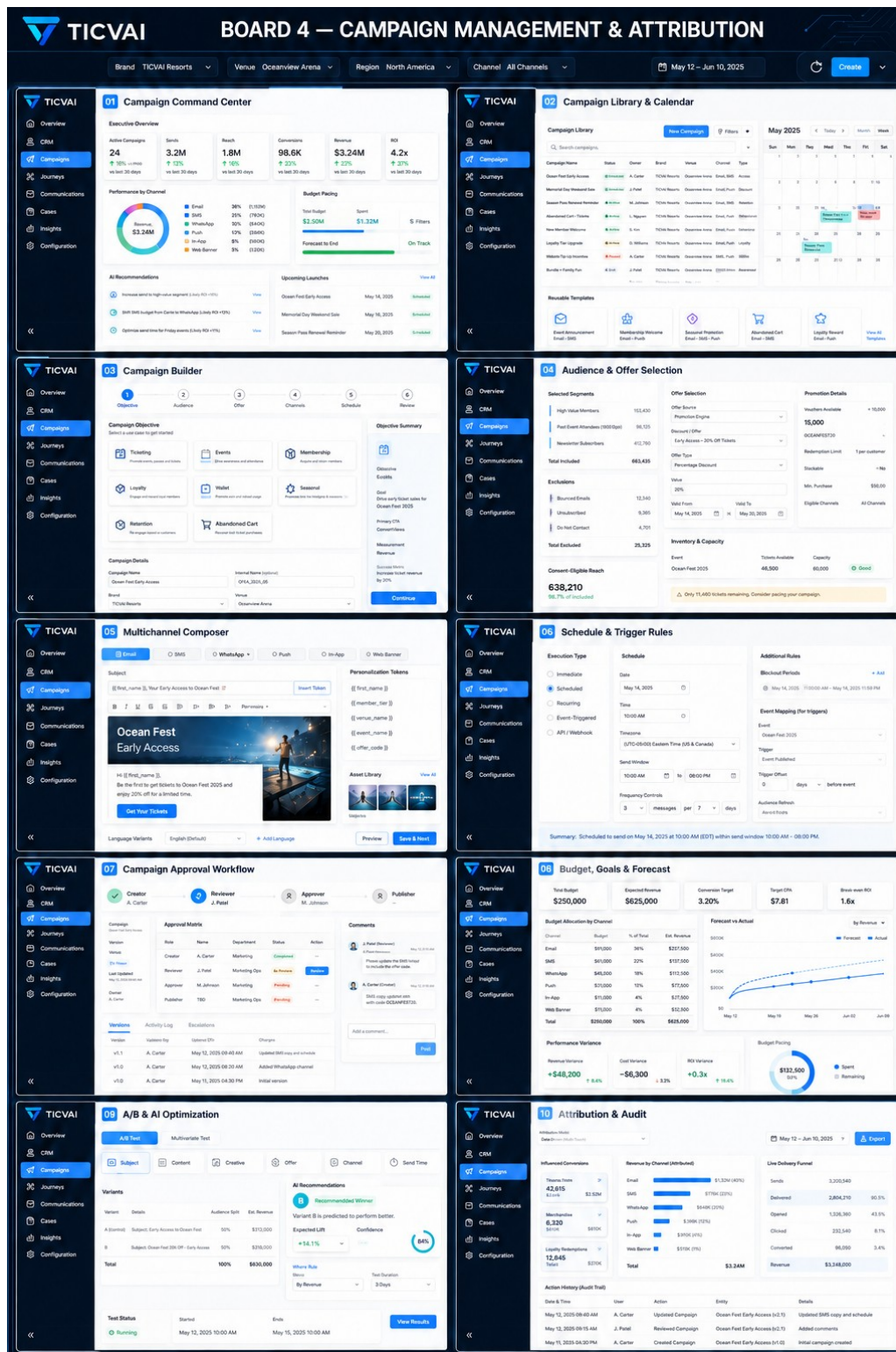


Figure 4. High-definition configuration board with all 10 screens.

Board 4 Scope of Work

Purpose. Manage the complete campaign lifecycle across ticketing, membership, loyalty, wallet and venue experiences, with approval, experimentation and defensible revenue attribution.

Source coverage. Matrix section 22.1 and attribution requirements in section 22.14

Operating workflow. Create from blank or template -> define objective -> select audience and offer -> compose channels -> schedule or trigger -> approve -> execute -> test and optimize -> attribute outcomes -> audit.

Mandatory integrations. Segmentation, consent, promotion engine, ticketing, capacity, membership, loyalty, wallet, assets, communications, CMS, mobile app, payments, analytics and finance.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Campaign Command Center

Purpose: Provide a real-time overview of campaign operations and commercial performance.

- Show active, scheduled, paused and completed campaigns with sends, reach, conversions, revenue, ROI and budget pacing.
- Compare email, SMS, WhatsApp, push, in-app, web and external-channel delivery and conversion.
- Display upcoming launches, approval bottlenecks, errors, capacity risks and underperforming campaigns.
- Provide explainable AI recommendations and drill-down to campaign, audience, channel and transaction detail.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Campaign Library & Calendar

Purpose: Organize campaigns, templates and execution schedules.

- Provide table, card and calendar views filtered by owner, brand, venue, region, objective, channel and status.
- Support reusable templates for ticketing, events, membership, loyalty, wallet, seasonal, retention and abandoned-cart campaigns.
- Allow clone, archive, compare, tag and controlled bulk actions while preserving lineage and versions.
- Show schedule conflicts, blackout periods, audience fatigue and related journey activity.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Campaign Builder

Purpose: Guide authorized users through complete campaign creation.

- Capture campaign name, objective, brand, venue, owner, products, commercial goal and measurement model.

- Support ticket, event, membership, loyalty, wallet, seasonal, retention, capacity and last-minute sales use cases.
- Provide a staged flow for objective, audience, offer, channels, schedule, review and activation.
- Validate required data, consent, inventory, offer, budget, channel and approval conditions before submission.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Audience & Offer Selection

Purpose: Connect governed audiences with eligible commercial offers.

- Select one or more segments, exclusions and suppression lists and show reachable, consent-eligible count by channel.
- Configure discounts, vouchers, promo codes, bundles, loyalty rewards, wallet bonuses and membership benefits through the Promotion Engine.
- Validate eligibility, stackability, redemption limits, dates, capacity, ticket inventory and product availability.
- Estimate exposure, cost, revenue and margin before proceeding.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Multichannel Composer

Purpose: Create personalized content for every selected communication channel.

- Provide channel-specific email, SMS, WhatsApp, push, in-app and website-banner editors and previews.
- Insert approved personalization tokens, products, events, tickets, QR codes, vouchers, payment links and media assets.
- Support brand and language variants, accessibility checks, link validation and character/channel constraints.
- Maintain common campaign intent while allowing controlled channel variations and version history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Schedule & Trigger Rules

Purpose: Configure when and under what conditions a campaign executes.

- Support immediate, scheduled, recurring, event-triggered and API/webhook execution.
- Configure timezone, send windows, blackout dates, frequency caps, re-entry, expiry and audience refresh timing.
- Map events such as purchase, expiry, birthday, milestone, top-up, first visit and abandoned cart.
- Provide schedule summary, conflict detection and pre-flight audience/inventory revalidation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Campaign Approval Workflow

Purpose: Enforce configurable review and publishing governance.

- Support creator, reviewer, approver and publisher stages with role, department, threshold and business-unit rules.
- Route content, offer, budget, legal, brand and data-privacy approvals independently where required.
- Capture comments, requested changes, resubmission, delegation, escalation and segregation-of-duties checks.
- Lock the approved version and audit every decision and subsequent change.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 08 - Budget, Goals & Forecast

Purpose: Plan campaign investment and expected commercial results.

- Capture total and channel budgets, expected revenue, conversion, CPA, margin, break-even and strategic goals.
- Estimate provider cost, offer cost, reward liability and capacity implications by channel and audience.
- Compare forecast, actual and committed spend and flag variance or overspend risk.
- Support approval thresholds, finance integration and controlled adjustment with reason and audit.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 09 - A/B & AI Optimization

Purpose: Test campaign alternatives and apply AI recommendations safely.

- Configure A/B or multivariate variants for subject, content, creative, offer, channel and send time.
- Define audience split, sample size, duration, success metric, statistical confidence and winner rule.
- Show AI recommendations, expected uplift, confidence, factors and constraints and require human approval.
- Prevent conflicting tests and preserve assignments, results, model/version and decisions for audit.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 10 - Attribution & Audit

Purpose: Measure influenced outcomes and preserve a complete campaign record.

- Provide first, last, linear, position-based and configurable multi-touch attribution across channels.
- Attribute ticket, membership, loyalty, wallet, reservation and other revenue using defined windows and identity rules.
- Display delivery funnel, conversion, revenue, ROI, profitability, channel contribution and unattributed outcomes.
- Record creation, edits, approvals, execution, pauses, variants, results, exports and user/AI actions.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 5 - Marketing Automation & Customer Journeys

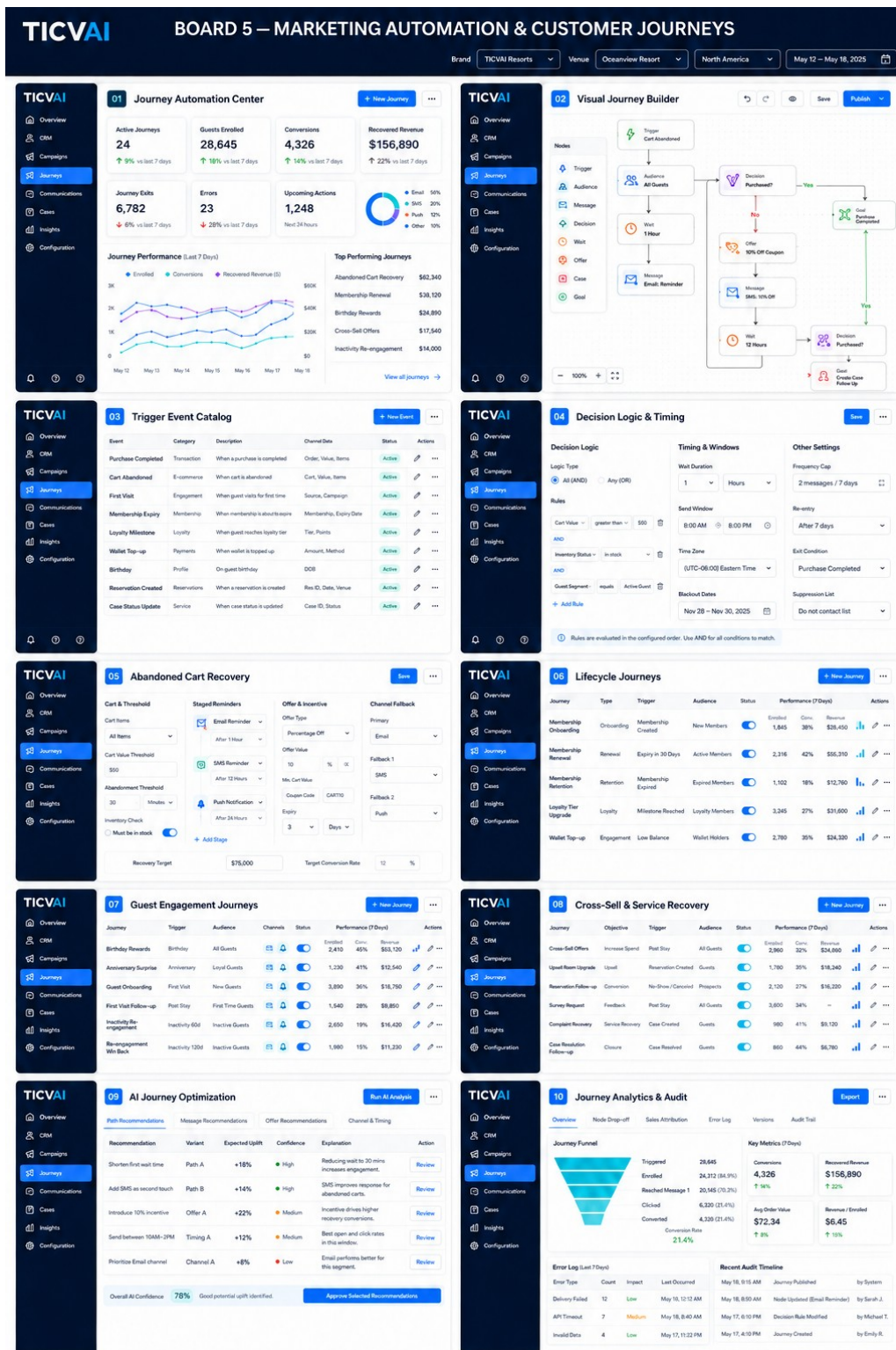


Figure 5. High-definition configuration board with all 10 screens.

Board 5 Scope of Work

Purpose. Configure event-driven, multi-step customer journeys that coordinate engagement, commerce and service recovery across channels and lifecycle stages.

Source coverage. Journey-automation requirements following matrix section 22.3.10, plus event-triggered campaign requirements

Operating workflow. Choose trigger and audience -> design path -> configure decisions, waits and actions -> validate -> approve and publish -> enroll guests -> monitor nodes and outcomes -> optimize and audit.

Mandatory integrations. Event bus, Customer Master, segmentation, campaigns, communications, promotion, ticketing, membership, loyalty, wallet, reservations, surveys, cases, payments and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Journey Automation Center

Purpose: Monitor the health and value of automated customer journeys.

- Show active journeys, enrolled guests, conversions, recovered revenue, exits, errors and upcoming actions.
- Compare journey performance by objective, lifecycle, brand, venue, audience, channel and owner.
- Surface paused nodes, delivery failures, SLA risk and data or consent problems requiring intervention.
- Provide drill-down and explainable optimization opportunities without automatically changing live flows.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Visual Journey Builder

Purpose: Create multi-step journeys through a visual no-code canvas.

- Provide trigger, audience, decision, wait, message, offer, case, task, webhook and goal nodes.
- Support branches, merges, reusable subflows, annotations, zoom, validation and realistic test execution.
- Show node configuration, eligible count, dependencies and errors without leaving the canvas.
- Version drafts, compare changes and require approval before a published journey is replaced.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Trigger Event Catalog

Purpose: Govern the events that can start, update or stop a journey.

- Configure purchase, cart abandonment, first visit, membership expiry, loyalty milestone, wallet top-up, birthday, reservation and case events.

- Define source, event schema, required attributes, deduplication, freshness, eligibility and sample payload.
- Support API/webhook events and scheduled conditions with authentication and retry controls.
- Version and test event definitions and display consuming journeys before deactivation or change.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 04 - Decision Logic & Timing

Purpose: Control journey branching, timing and re-entry.

- Build nested AND/OR conditions using guest, event, segment, transaction, product, channel and engagement data.
- Configure waits, dates, timezones, send windows, blackout periods, frequency limits and expiry.
- Define re-entry, concurrency, suppression, exit and goal conditions and behavior when data is missing.
- Simulate paths for representative guests and block publication when a branch has no safe outcome.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 05 - Abandoned Cart Recovery

Purpose: Configure automated recovery of incomplete ticketing and commerce transactions.

- Define abandonment event, cart-value threshold, product/inventory checks, eligibility and recovery window.
- Configure staged reminders across email, SMS, WhatsApp and push with fallback and frequency limits.
- Apply approved incentives through the Promotion Engine and prevent discount leakage or expired inventory offers.
- Track recovered cart, revenue, margin, attribution and exit when purchase or cancellation occurs.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 06 - Lifecycle Journeys

Purpose: Manage reusable membership, loyalty and wallet lifecycle automations.

- Provide onboarding, benefit-awareness, renewal, expiry, lapse, freeze and upgrade membership journeys.
- Provide loyalty enrollment, tier movement, milestone, reward availability and points-expiry journeys.
- Provide wallet onboarding, top-up, low-balance, dormancy and bonus-credit journeys.
- Configure eligibility, channels, goals, stop conditions, ownership, versions and lifecycle-specific KPIs.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Screen 07 - Guest Engagement Journeys

Purpose: Configure recurring guest relationship journeys.

- Support birthdays, anniversaries, welcome/onboarding, first-visit follow-up, inactivity and re-engagement.
- Personalize by profile, language, interests, visit history, loyalty, membership and preferred channel.
- Apply quiet hours, consent, frequency and household/minor rules before every communication.
- Measure engagement, conversion, return visits, retention and incremental value by journey.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Cross-Sell & Service Recovery

Purpose: Coordinate revenue-growth and recovery actions after guest events.

- Recommend and automate eligible cross-sell or upsell offers after purchase, booking or visit.
- Trigger reservation follow-up, survey request, complaint recovery and post-case resolution journeys.
- Create cases, tasks, refunds, vouchers, loyalty points or wallet credit subject to approval limits.
- Stop or alter a journey when sentiment, case status, refund, opt-out or service outcome changes.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - AI Journey Optimization

Purpose: Use explainable AI to improve paths, content, offers and timing.

- Recommend branch order, wait duration, channel, message, offer and send time with expected uplift.
- Display supporting evidence, confidence, affected audience, risks and policy constraints.
- Allow scenario comparison, selective approval and rollback and prevent AI from editing live journeys directly.
- Monitor applied recommendation results and retain model/version, reviewer feedback and realized impact.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Journey Analytics & Audit

Purpose: Measure journey behavior at node and end-goal level.

- Show triggered, eligible, enrolled, reached, clicked, converted, exited and failed stages.
- Analyze node drop-off, branch performance, time-to-conversion, revenue, attribution and cost.
- Provide error log, delivery trace, suppression reason, version comparison and guest-level path where authorized.
- Audit creation, edits, approvals, publication, pauses, AI recommendations and administrative actions.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 6 - Newsletter, Notifications & Message Delivery

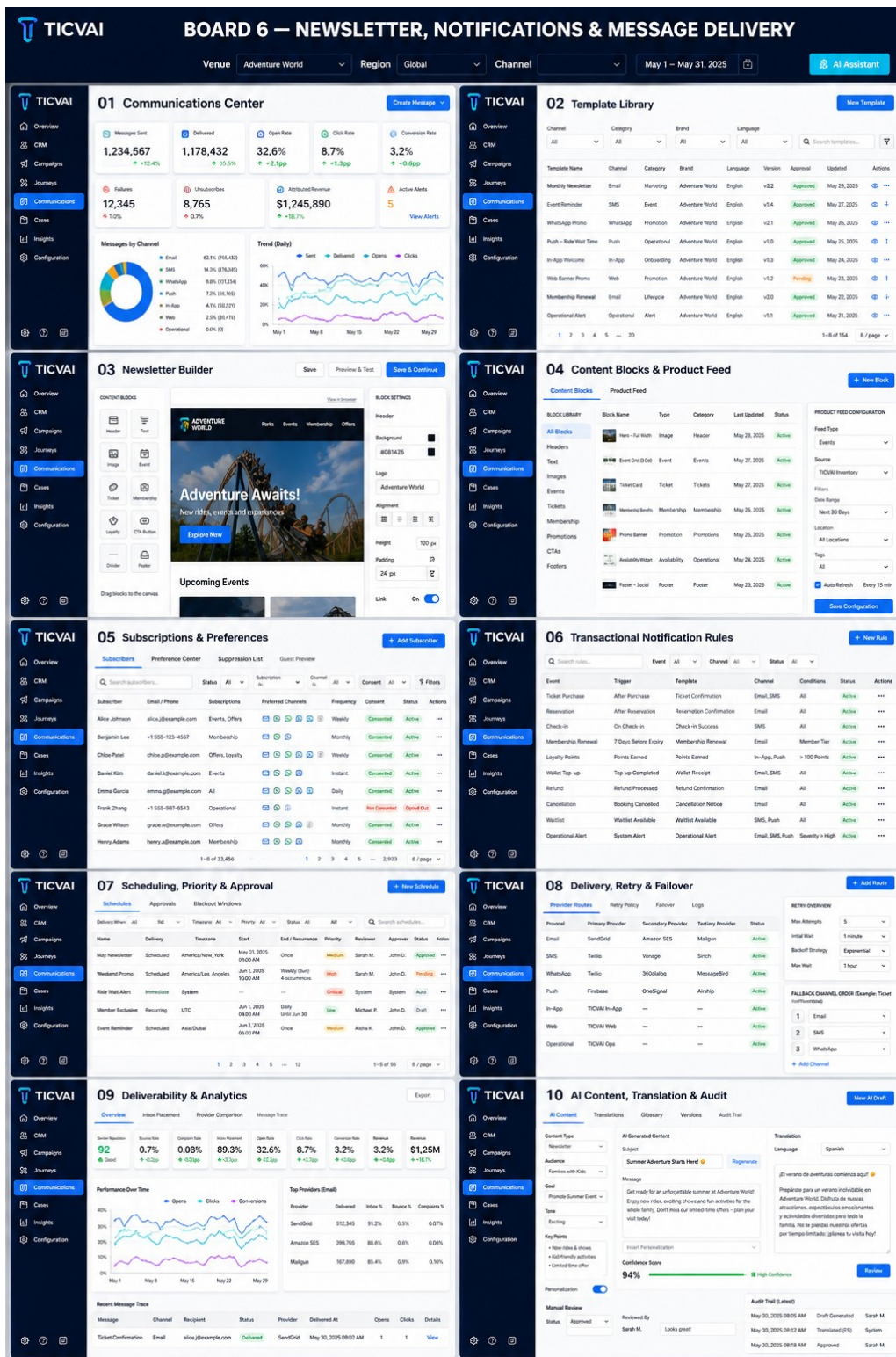


Figure 6. High-definition configuration board with all 10 screens.

Board 6 Scope of Work

Purpose. Provide one governed communication engine for marketing newsletters, transactional notifications, operational alerts and rich personalized messages across all supported channels.

Source coverage. Matrix sections 22.4 - Newsletter & Email Marketing and 22.9 - Notifications & Messaging

Operating workflow. Create or select template -> compose or bind event data -> apply personalization and language -> schedule/approve -> route to provider -> retry or fail over -> record outcome -> analyze and audit.

Mandatory integrations. Consent and preferences, Customer Master, ticketing, reservations, membership, loyalty, wallet, campaigns, journeys, CMS, provider gateways, payments, mobile/web apps and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Communications Center

Purpose: Monitor outbound communication volume, delivery and engagement.

- Show messages sent, delivered, failed, opened, clicked, converted and unsubscribed with attributed revenue.
- Compare email, SMS, WhatsApp, push, in-app, web and operational channels and providers.
- Display active alerts for provider outage, high bounce, consent error, queue delay or retry exhaustion.
- Filter and drill down by tenant, brand, venue, region, template, category, channel and date.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Template Library

Purpose: Centrally manage reusable message templates.

- Organize marketing, ticketing, reservation, membership, loyalty, wallet and operational templates by channel and purpose.
- Maintain brand, venue, language, version, owner, approval, validity and active/inactive status.
- Preview test data, validate required tokens, links, attachments and channel-specific constraints.
- Clone, archive and compare templates while preserving lineage and complete change history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Newsletter Builder

Purpose: Create responsive, branded newsletters without coding.

- Provide drag-and-drop header, text, image, event, ticket, membership, loyalty, CTA, divider and footer blocks.

- Support personalization, dynamic content, reusable templates, language variants and desktop/mobile preview.
- Validate accessibility, links, unsubscribe content, sender details and deliverability before test or approval.
- Support draft, test send, review, approve, schedule, clone and version-management workflows.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Content Blocks & Product Feed

Purpose: Connect communications to reusable content and live products.

- Maintain approved content blocks for banners, copy, events, tickets, memberships, loyalty, promotions, CTAs and footers.
- Configure live feeds from events, products, ticket inventory, capacity, pricing, promotions and recommendations.
- Define filters, sorting, fallback content, refresh interval, expiration and behavior when inventory is unavailable.
- Track which campaigns and templates use each block before editing or deactivation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Subscriptions & Preferences

Purpose: Manage newsletter subscriptions and messaging eligibility.

- Configure subscription categories, topics, channels, frequency, consent mapping, suppression and do-not-contact status.
- Search subscribers and view preferred channels, verification, consent source, status and last change.
- Support guest self-service and authorized administration while preventing forced marketing opt-in.
- Synchronize changes immediately to campaign, journey, newsletter and delivery services and audit them.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Transactional Notification Rules

Purpose: Map operational events to the correct notification behavior.

- Configure rules for purchase, reservation, check-in, membership, loyalty, wallet, refund, cancellation, waitlist and operational events.
- Select template, channel, recipients, timing, priority, conditions, attachments and personalization mapping.
- Differentiate mandatory transactional notices from optional marketing and apply the correct consent policy.
- Test rules with sample events and prevent conflicting or duplicate notifications.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Scheduling, Priority & Approval

Purpose: Control message timing, urgency and governance.

- Support immediate, scheduled, recurring and event-relative delivery with timezone, windows and blackout periods.
- Configure Critical, High, Medium and Low priority, queue treatment, escalation and delivery deadline.
- Route selected templates and schedules through reviewer and approver stages with delegation and comments.
- Provide a final audience, content, channel, cost and policy pre-flight before activation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Delivery, Retry & Failover

Purpose: Configure reliable routing across communication providers and channels.

- Maintain primary, secondary and tertiary providers per channel, tenant, region and message category.
- Define retry count, delay, backoff, retryable errors, expiration and dead-letter handling.
- Configure channel failover order while rechecking consent, content compatibility and urgency.
- Provide message trace, provider response, failure reason, manual replay controls and audit history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Deliverability & Analytics

Purpose: Monitor technical delivery quality and commercial outcomes.

- Report sender reputation, inbox placement, hard/soft bounce, complaints, blocks, opens, clicks and conversions.
- Compare providers, domains, channels, templates, languages, brands and regions over time.
- Trace individual messages from generation to provider response and guest engagement where authorized.
- Alert on thresholds and attribute revenue using the shared marketing attribution framework.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - AI Content, Translation & Audit

Purpose: Use governed AI for communication creation and localization.

- Generate subject lines, titles, summaries and message variants by objective, tone, audience and channel.
- Translate content using approved languages, terminology glossary, brand rules and protected placeholders.

- Display confidence and changes, require manual review where configured and prevent unsupported claims.
- Retain prompt/context classification, model/version, output, edits, approver and publication audit evidence.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 7 - Omnichannel Inbox, AI Chatbot & Agent Workspace

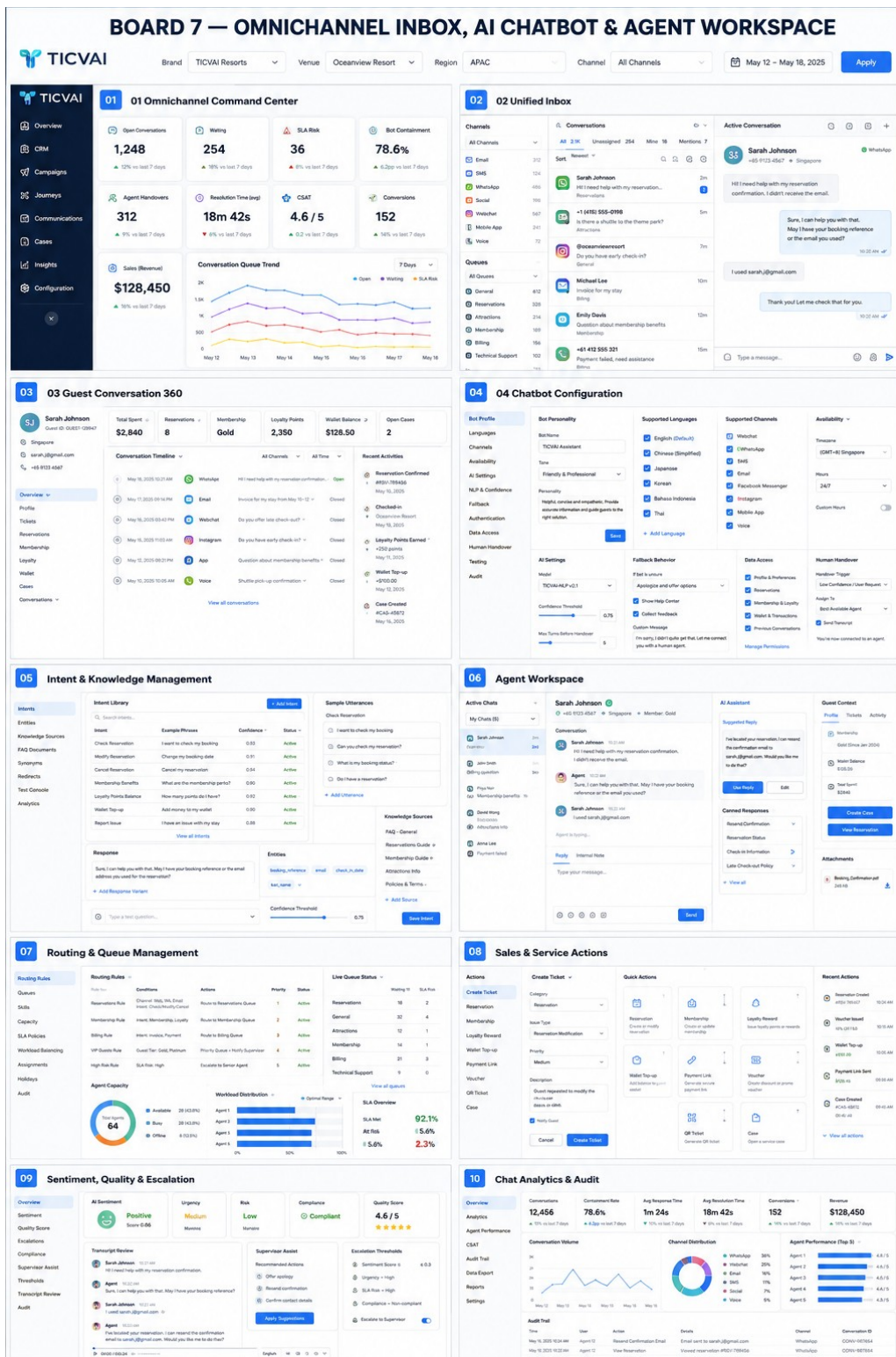


Figure 7. High-definition configuration board with all 10 screens.

Board 7 Scope of Work

Purpose. Provide one service-and-sales workspace for guest conversations across channels, supported by a governed AI assistant and live-agent handover.

Source coverage. Matrix section 22.8 - Omnichannel Communications & AI Chatbot

Operating workflow. Receive and identify conversation -> classify intent and sentiment -> answer through AI or route to queue -> hand over with full context -> perform ticketing/service/payment actions -> resolve -> analyze and audit.

Mandatory integrations. Customer 360, ticketing, reservations, membership, loyalty, wallet, payment links, campaigns, promotions, cases, knowledge base, communication providers, mobile app and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Omnichannel Command Center

Purpose: Monitor conversation demand, service level and commercial contribution.

- Show open, waiting, SLA-risk and escalated conversations, bot containment, agent handover, response and resolution time.
- Display CSAT, conversions, revenue and queue/channel/venue distribution with trend and capacity indicators.
- Surface outages, knowledge gaps, negative sentiment spikes and overloaded queues.
- Provide role-specific drill-down for supervisors, agents, sales, support and administrators.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Unified Inbox

Purpose: Consolidate all guest conversations into one manageable queue.

- Ingest email, SMS, WhatsApp, social, webchat, mobile-app chat and supported voice/messaging channels.
- Provide queue and channel filters, assignment, priority, unread state, SLA clock, search and saved views.
- Display the active conversation, rich messages, attachments, delivery/read status and internal notes.
- Prevent duplicate threads through identity and conversation correlation and retain the source channel.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Guest Conversation 360

Purpose: Combine conversation history with the identified guest's CRM context.

- Resolve the guest and display profile, value, tickets, reservations, membership, loyalty, wallet and open cases.

- Show all prior conversations across channels in chronological order with intent, outcome and agent/bot ownership.
- Allow authorized users to open linked records without losing the current conversation.
- Mask or restrict sensitive information and show confidence when identity is uncertain.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - AI Chatbot Configuration

Purpose: Configure the AI assistant's identity, scope and operating controls.

- Set bot name, personality, tone, supported languages, channels, operating hours and brand/venue context.
- Configure confidence thresholds, fallback, authentication, guest-data access and maximum automated turns.
- Define permitted sales/service actions, prohibited topics, safe responses and human-handover conditions.
- Version, approve, test and audit configuration before deployment to any channel.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Intent & Knowledge Management

Purpose: Manage what the assistant understands and the information it may use.

- Maintain intents, example phrases, entities, synonyms, redirects, confidence and response variants.
- Connect approved FAQs, product data, policies, manuals and structured knowledge with freshness and ownership.
- Provide test console, expected result, source citation, ambiguity handling and low-confidence review queue.
- Track knowledge consumers, sync status, changes and answer-quality feedback.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Agent Workspace

Purpose: Give live agents the context and tools required to resolve conversations efficiently.

- Provide active conversation, transcript, guest context, AI reply suggestion, canned responses and translation.
- Support rich messages, attachments, internal notes, tags, tasks, transfer and supervisor assistance.
- Expose authorized ticket, reservation, membership, loyalty, wallet and case actions without switching applications.
- Track agent presence, typing, ownership, response time, action outcomes and complete transcript history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Routing & Queue Management

Purpose: Route work to the correct team and balance workload.

- Create rules by intent, skill, department, venue, attraction, language, guest tier, priority and SLA risk.
- Configure queues, capacity, operating hours, overflow, assignment method, workload limits and fallback.
- Provide live queue counts, wait time, agent utilization, reassignment, pickup and supervisor override.
- Record the evaluated rule, routing reason, transfers and any manual intervention.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Sales & Service Actions

Purpose: Allow agents and AI to perform governed transactional actions.

- Create or modify tickets and reservations and view capacity, eligibility, price and policy before confirmation.
- Manage membership information, loyalty rewards and wallet top-ups within configured authority.
- Generate secure payment links, vouchers, QR tickets, forms and rich messages and confirm delivery.
- Create and escalate cases and require authentication, approval or live-agent control for sensitive actions.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Sentiment, Quality & Escalation

Purpose: Detect conversation risk and support consistent service quality.

- Analyze sentiment, emotion, urgency, intent confidence, compliance and quality score in near real time.
- Configure thresholds for supervisor alert, priority change, queue transfer, case creation or live-agent takeover.
- Provide transcript review, AI rationale, quality checklist, coaching notes and appeal/override controls.
- Monitor false positives, bias, model drift and outcomes and audit automated and human escalations.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Chat Analytics & Audit

Purpose: Measure channel, chatbot and agent performance and preserve conversation evidence.

- Report volume, response, resolution, abandonment, containment, handover, transfer, conversion, revenue and CSAT.
- Compare channel, queue, venue, intent, language, bot version, agent and time period.

- Provide conversation drill-down, action trace, AI/human ownership, case links and payment/sales outcome.
- Audit messages, transfers, AI actions, escalations, administrative changes and data access.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 8 - Case Management, SLA & Service Recovery

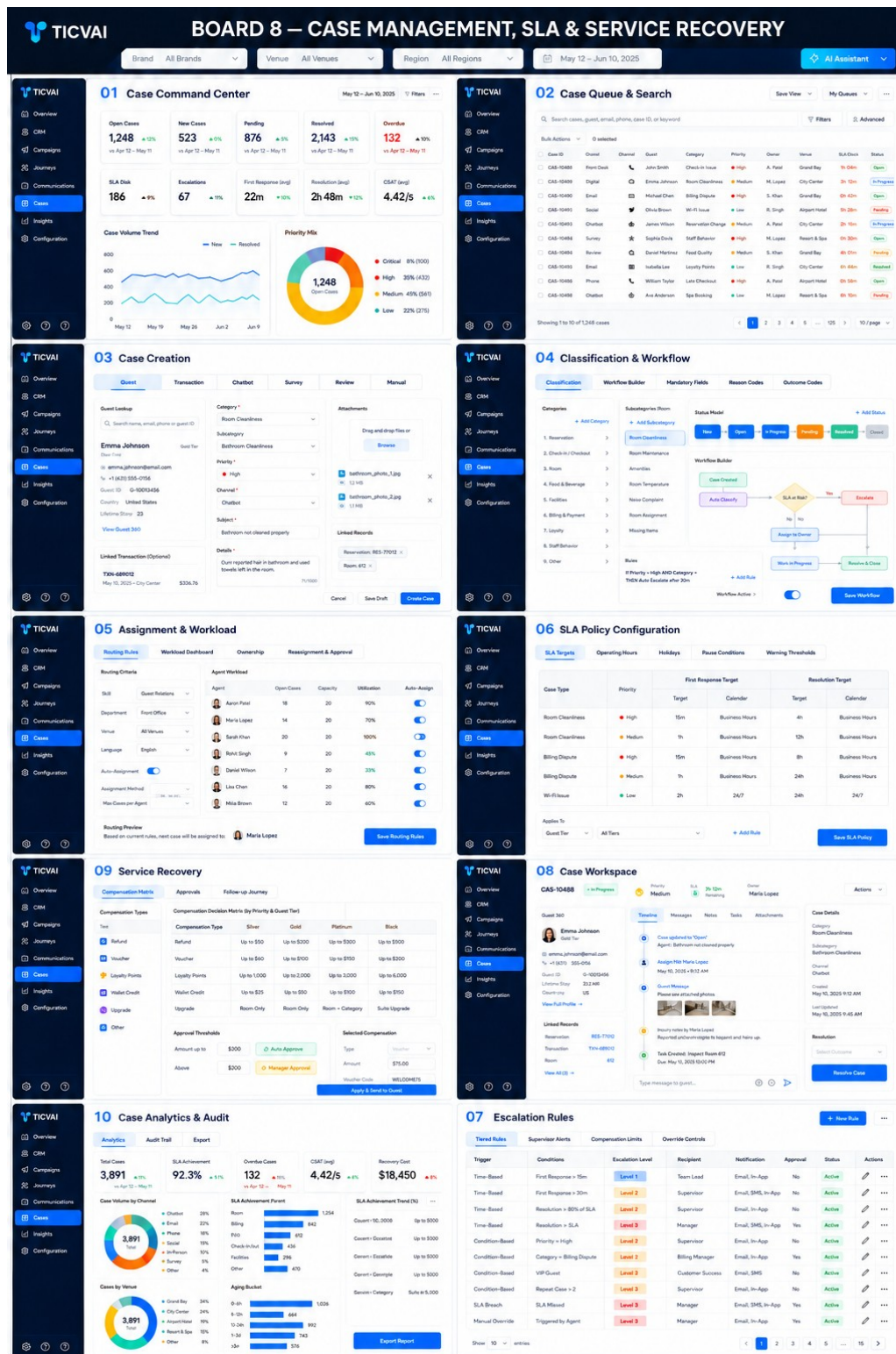


Figure 8. High-definition configuration board with all 10 screens.

Board 8 Scope of Work

Purpose. Manage complaints, incidents, requests and service recovery from intake through SLA-controlled resolution and measurable guest outcome.

Source coverage. First ten requirements in matrix section 22.3 - Case Management & SLA

Operating workflow. Create or receive case -> classify -> assign -> work and communicate -> monitor SLA -> escalate -> approve recovery -> resolve -> measure cause, cost and satisfaction -> audit.

Mandatory integrations. Customer 360, omnichannel inbox, surveys, reviews, ticketing, reservations, payments/refunds, promotion, loyalty, wallet, workflow approvals, employee tasks, notifications and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Case Command Center

Purpose: Provide a role-specific overview of service operations.

- Show new, open, pending, resolved, overdue and SLA-risk cases with escalations, response time, resolution time and CSAT.
- Visualize volume, priority, category, channel, venue, owner, aging and trend.
- Surface major incidents, repeated issues, overloaded queues and high-value guests requiring attention.
- Provide drill-down to saved queues and allow supervisors to take governed corrective action.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Case Queue & Search

Purpose: Provide a complete, searchable work queue for service cases.

- Search by case, guest, contact, transaction, ticket, booking, keyword or external reference.
- Filter by category, subcategory, priority, channel, venue, status, owner, SLA, date and guest tier.
- Support saved views, configurable columns, controlled bulk assignment/status actions and export.
- Display live SLA clocks and role-based action availability without exposing restricted guest data.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Case Creation

Purpose: Create a complete case from any service touchpoint.

- Create manually or from chatbot, conversation, survey, review, transaction, incident or API event.
- Capture guest, category, priority, channel, venue, subject, details, attachments and linked records.
- Apply defaults, mandatory fields, duplicate detection, suggested classification and SLA preview.
- Allow draft and submit workflows and record source, creator, timestamp and initial evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Classification & Workflow

Purpose: Configure case taxonomies and lifecycle states.

- Maintain categories, subcategories, reason codes, outcomes, severity, priority and applicable business units.
- Configure status model, stage transitions, mandatory fields, validation and conditional steps.
- Use a visual workflow for assignment, work, approval, escalation, resolution and closure.
- Version and approve configurations and show affected open cases before activating changes.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Assignment & Workload

Purpose: Assign cases to the best available team or agent.

- Route by skill, department, venue, language, category, guest tier, priority and operating hours.
- Configure round robin, least load, ownership continuity, named team and manual assignment methods.
- Display capacity, utilization, open workload, absence and assignment recommendation.
- Require approval for restricted reassignment and retain assignment reason and history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - SLA Policy Configuration

Purpose: Define measurable service commitments and pause rules.

- Configure first-response and resolution targets by case type, priority, venue, channel and guest tier.
- Apply operating calendars, holidays, 24/7 rules, pause conditions, dependency states and warning thresholds.
- Define target hierarchy, precedence, exceptions and behavior when multiple SLAs apply.
- Simulate policies before activation and version, approve and audit every change.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Escalation Rules

Purpose: Automate tiered escalation before or after service risk materializes.

- Create time-based and condition-based triggers for response, resolution, priority, sentiment, repeat contact and compliance risk.
- Define escalation level, recipients, notification channel, required action, approval and acknowledgement deadline.
- Configure supervisor, manager, specialist and executive paths, compensation limits and override controls.
- Prevent escalation loops and record trigger, evaluated conditions, recipients, actions and closure.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Case Workspace

Purpose: Provide one place to investigate, communicate and resolve a case.

- Display guest 360, case details, SLA, timeline, messages, notes, tasks, attachments and linked transactions.
- Support email, SMS, WhatsApp and chat responses using approved templates and complete history.
- Allow assignment, status, priority, task, approval, related-case and resolution actions by authority.
- Require reason and evidence for sensitive changes and preserve immutable history after closure.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Service Recovery

Purpose: Configure and approve fair, consistent recovery actions.

- Maintain compensation matrices by issue, severity, guest tier, value and business unit.
- Support refund, voucher, loyalty points, wallet credit, upgrade, replacement and follow-up journey actions.
- Apply auto-approval and manager-approval thresholds, budget limits, fraud checks and segregation of duties.
- Track recovery cost, delivery, redemption, guest response and effect on satisfaction and retention.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Case Analytics & Audit

Purpose: Measure service quality, root causes and operational accountability.

- Report case volume, SLA achievement, aging, backlog, escalation, resolution, CSAT and recovery cost.
- Analyze category, root cause, channel, venue, product, guest tier, team and agent performance.
- Track repeat cases, lost revenue, compensation, recovery effectiveness and systemic improvement actions.
- Provide complete audit, secured export and traceability from reported issue to final resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Board 9 - Surveys, Reviews & Voice of Customer

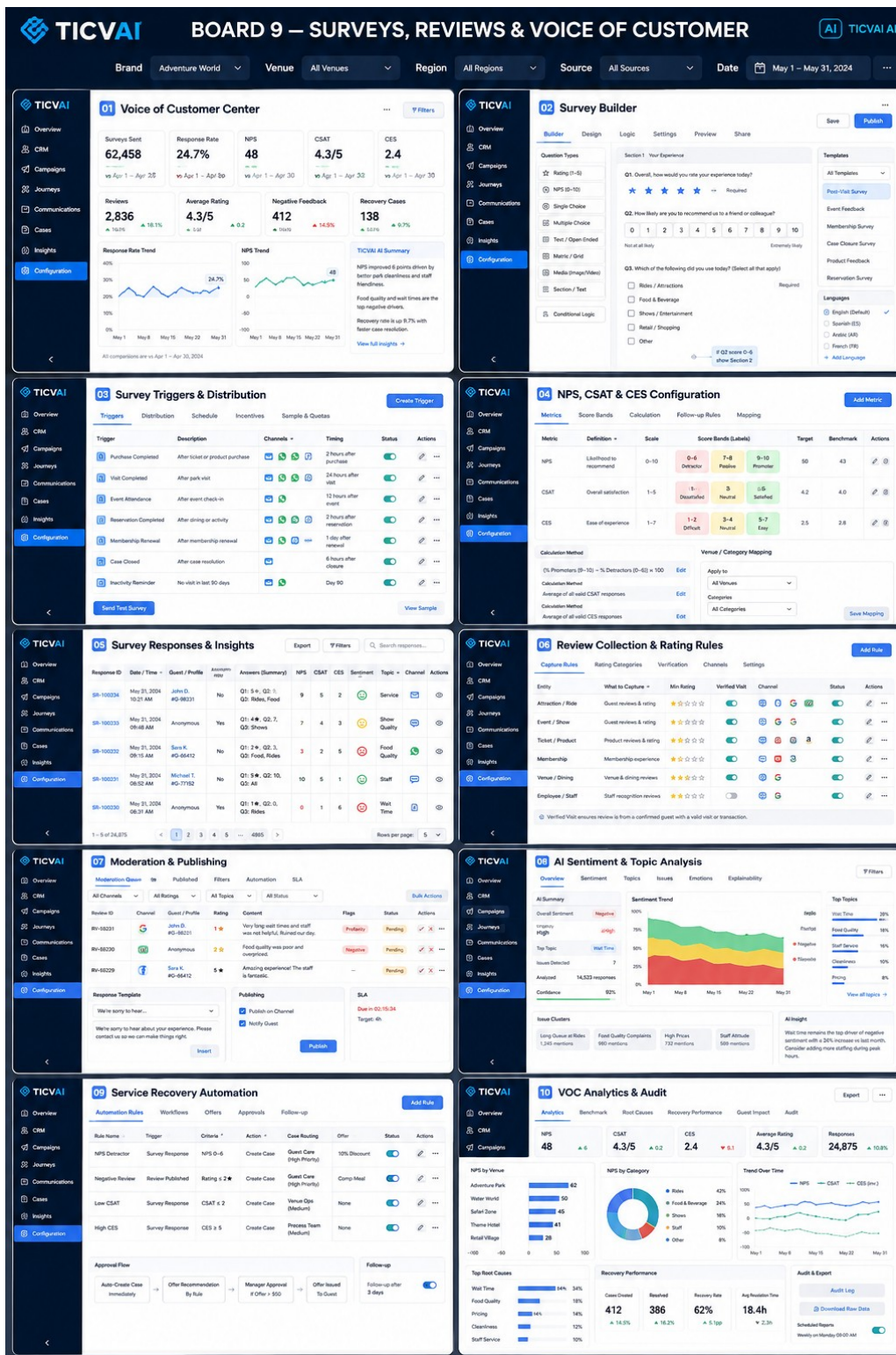


Figure 9. High-definition configuration board with all 10 screens.

Board 9 Scope of Work

Purpose. Collect, analyze and act on structured and unstructured guest feedback across visits, products, events and service interactions.

Source coverage. Matrix sections 22.5 - Reviews & Ratings and 22.7 - Surveys & Feedback

Operating workflow. Design survey/review rules -> trigger and distribute -> collect response -> calculate metrics -> moderate/publish -> analyze sentiment and topics -> create recovery action -> monitor outcomes and audit.

Mandatory integrations. Customer 360, ticketing/visit validation, reservations, membership, loyalty, communications, campaigns, journeys, cases, public review channels, mobile app and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Voice of Customer Center

Purpose: Provide a consolidated view of guest feedback and recovery performance.

- Show surveys sent, response rate, NPS, CSAT, CES, review count, average rating and negative-feedback volume.
- Visualize metric, rating, sentiment, topic and recovery trends by brand, venue, product, event and channel.
- Surface urgent detractors, repeated issues, moderation backlog and recovery cases at risk.
- Provide AI summaries with links to underlying responses and explainable topic evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Survey Builder

Purpose: Create multilingual, accessible surveys without development.

- Provide rating, NPS, single choice, multiple choice, text, matrix, date, media and information-section elements.
- Support question library, templates, required fields, validation, page/section layout and conditional branching.
- Configure identified or anonymous response, consent statement, incentive and completion behavior.
- Preview by channel and language and version, approve and test before publication.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Survey Triggers & Distribution

Purpose: Control when, where and to whom a survey is sent.

- Trigger after purchase, visit, event, reservation, membership interaction, case closure or inactivity.
- Distribute through email, SMS, WhatsApp, push, mobile app, web, kiosk and QR where supported.

- Configure delay, expiry, reminder, sample, quota, frequency cap, exclusions and incentive.
- Validate consent and eligibility and prevent duplicate or excessive survey requests.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - NPS, CSAT & CES Configuration

Purpose: Define standardized experience metrics and follow-up thresholds.

- Configure metric definition, scale, labels, score bands, calculation and rounding.
- Set targets, benchmarks, venue/category mappings and reporting periods.
- Define detractor, low-CSAT and high-effort follow-up actions and case priority.
- Version metric rules and prevent historical results from being silently recalculated.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Survey Responses & Insights

Purpose: Review individual responses and their calculated insight.

- Show respondent/profile or anonymous status, survey/version, answers, channel, timestamps and incentive outcome.
- Display NPS, CSAT, CES, sentiment, topics, urgency and linked guest/transaction where permitted.
- Filter, search, export, tag, assign follow-up and open a service-recovery case.
- Protect anonymity and sensitive free text and preserve the submitted response unchanged.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Review Collection & Rating Rules

Purpose: Configure verified review and rating capture.

- Define review entities including attraction, event, ticket, product, membership, venue, dining and employee/service.
- Configure overall and category ratings, minimum/maximum scale, required comments and media support.
- Validate verified visits or purchases, identity, submission window, duplicates and fraud/spam signals.
- Map capture channels and decide which reviews require moderation before publication.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Moderation & Publishing

Purpose: Review and publish guest feedback safely and consistently.

- Provide moderation queue with channel, rating, text, media, flags, sentiment, topic and SLA.
- Approve, reject, redact, escalate or request follow-up using reason codes and permission controls.
- Configure profanity, personal-data, fraud and policy checks and public response templates.

- Publish to approved TICVAI and external channels and retain original, moderated and published versions.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - AI Sentiment & Topic Analysis

Purpose: Analyze feedback at scale using explainable AI.

- Classify sentiment, emotion, urgency, topics, issue type and intent across survey and review text.
- Cluster recurring issues and summarize themes by venue, product, event, language and time period.
- Show confidence, supporting excerpts, model/version and correction controls for reviewers.
- Monitor quality, bias and drift and restrict use of sensitive data or unsupported inference.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Service Recovery Automation

Purpose: Turn negative feedback into governed service action.

- Create rules for detractors, low CSAT, high effort, negative review, urgent topic and repeated issue.
- Create and route a case, select priority, notify the owner and recommend an approved recovery offer.
- Require approval based on compensation amount, guest tier, issue severity and fraud risk.
- Schedule follow-up, capture resolution and measure recovery, rating change, satisfaction and retention.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - VOC Analytics & Audit

Purpose: Provide cross-source performance and governance reporting.

- Combine NPS, CSAT, CES, ratings, sentiment, topics and response rates without double-counting guests.
- Benchmark venues, products, events, categories, channels and periods and expose root-cause drivers.
- Measure moderation SLA, recovery volume, recovery success, compensation cost and guest impact.
- Audit survey/review setup, AI analysis, moderation, publishing, case creation, exports and administrative actions.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Board 10 - Gamification & Loyalty Engagement

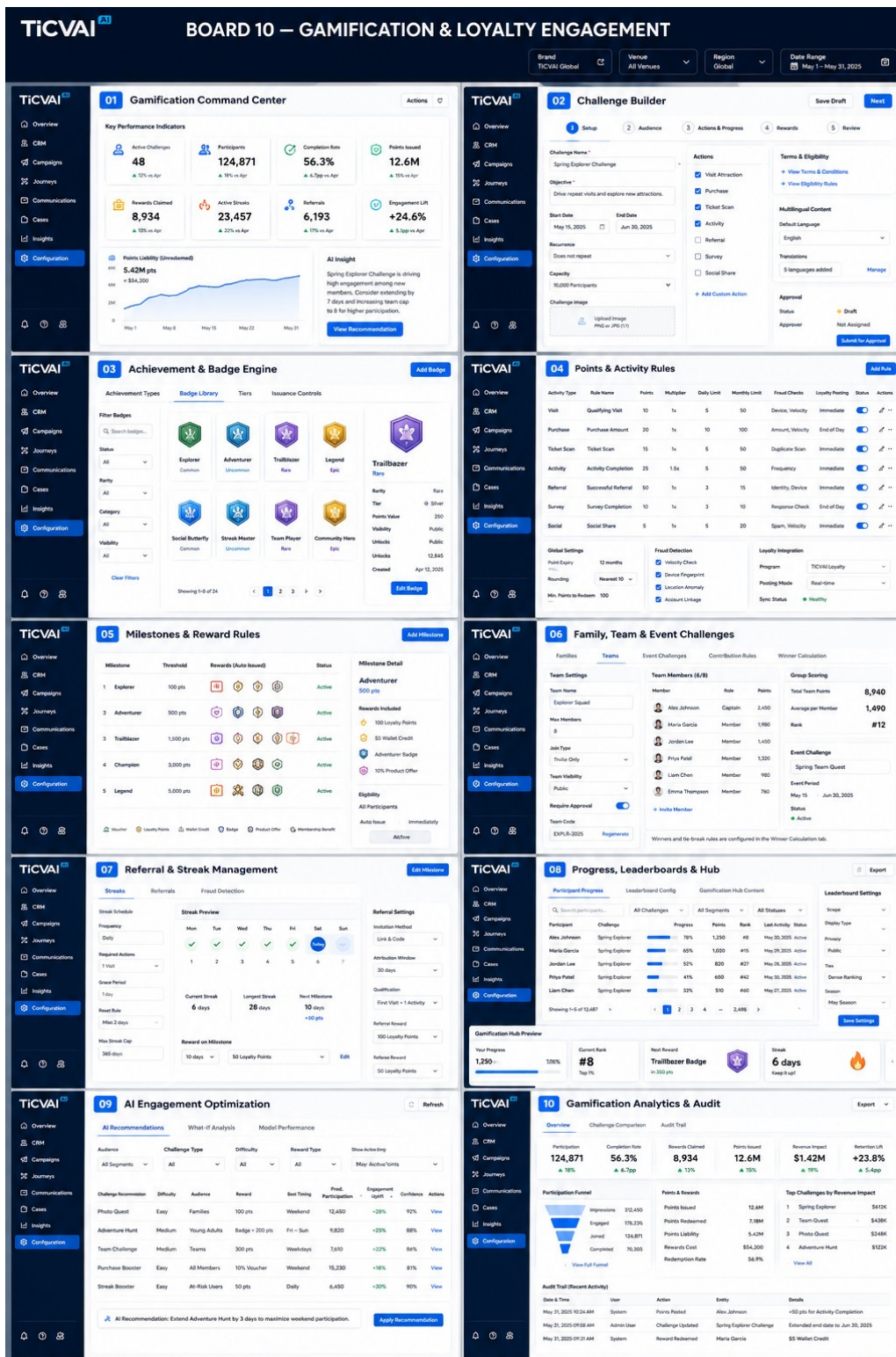


Figure 10. High-definition configuration board with all 10 screens.

Board 10 Scope of Work

Purpose. Configure challenges, achievements, progress and rewards that increase participation, visits and loyalty while controlling points liability and abuse.

Source coverage. Matrix section 22.6 - Gamification & Loyalty

Operating workflow. Define objective and audience -> configure activity and progress -> assign badge/points/reward -> approve and publish -> track participant progress -> fulfill rewards -> optimize -> analyze and audit.

Mandatory integrations. Customer 360, loyalty engine, membership, wallet, ticket scans, reservations, events, mobile app, campaigns, communications, referral tracking, promotion/reward marketplace and analytics.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Gamification Command Center

Purpose: Monitor engagement programs and their financial and behavioral impact.

- Show active challenges, participants, completion, points issued, rewards claimed, streaks and referrals.
- Report engagement lift, repeat visits, revenue impact and unredeemed points/reward liability.
- Compare by challenge, audience, venue, event, membership, team, season and period.
- Surface fraud anomalies, overspend, low participation and explainable AI opportunities.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Challenge Builder

Purpose: Create a governed challenge from objective through publication.

- Configure challenge name, objective, audience, actions, progress logic, start/end, recurrence and capacity.
- Define eligibility, terms, visibility, multilingual content, image, completion rule and reward.
- Support visit, purchase, scan, activity, referral, survey and approved social actions.
- Validate overlapping rules, reward budget, fraud controls, legal terms and approval before activation.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Achievement & Badge Engine

Purpose: Manage achievements and digital badges.

- Define achievement type, badge name, tier, rarity, unlock criteria, points value, visibility and validity.
- Maintain an approved visual badge library and multilingual name/description and accessible alt text.

- Configure automatic or manual issuance, revocation, duplicate prevention and profile/mobile display.
- Track active challenges and rewards using each achievement before editing or retiring it.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Points & Activity Rules

Purpose: Translate qualifying guest actions into controlled loyalty points.

- Configure rules for visits, purchases, ticket scans, activities, referrals, surveys and other approved events.
- Set base points, multiplier, daily/monthly limit, validity, rounding and posting timing.
- Apply device, account, velocity, location and transaction fraud checks and exception review.
- Post through the shared Loyalty Engine and preserve source event, rule/version and adjustment history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Milestones & Reward Rules

Purpose: Reward cumulative progress consistently.

- Define thresholds, tiers and immediate or delayed reward fulfillment.
- Support loyalty points, wallet credit, voucher, product offer, badge and membership benefit rewards.
- Configure eligibility, inventory, validity, redemption limit, substitution and approval threshold.
- Forecast reward cost/liability and prevent issuance when funding, inventory or eligibility fails.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Family, Team & Event Challenges

Purpose: Support collaborative and competitive engagement formats.

- Configure family/household, team, group and event membership and joining rules.
- Define individual versus shared contribution, group scoring, maximum members and visibility.
- Configure ranking, winner calculation, tie-break, prize allocation and guardian rules for minors.
- Show participant roster, progress, status, exceptions and auditable membership changes.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Referral & Streak Management

Purpose: Configure repeat-behavior and referral mechanics.

- Define streak frequency, qualifying event, grace period, reset, maximum, milestone and reward.
- Configure referral code/link, attribution window, qualifying action, inviter/invitee reward and limits.
- Apply identity, account, device, payment, velocity and collusion fraud checks.

- Track invitation, qualification, reward posting, reversal and dispute history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Progress, Leaderboards & Hub

Purpose: Control how participants see progress and rankings.

- Show progress, points, milestones, next reward, streak and challenge status by participant.
- Configure leaderboard metric, scope, season, filters, dense ranking, privacy display and opt-out.
- Manage mobile gamification-hub content, featured challenges, quick actions and deep links.
- Provide near-real-time updates and resolve corrections without losing historical ranking evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - AI Engagement Optimization

Purpose: Recommend relevant, sustainable engagement programs.

- Recommend challenge type, audience, difficulty, actions, reward and timing using historical behavior.
- Predict participation, completion, engagement uplift, cost and liability with confidence and factors.
- Support what-if comparison and require approval before a suggestion becomes a challenge or rule.
- Monitor applied recommendation results and avoid manipulative, discriminatory or age-inappropriate designs.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Gamification Analytics & Audit

Purpose: Measure challenge performance, cost and long-term effect.

- Report participant funnel, completion, points, rewards, redemption, cost, revenue and repeat visits.
- Compare challenges, audiences, venues, events, seasons, reward types and membership groups.
- Measure retention and engagement lift against appropriate baselines and show points/reward liability.
- Audit definitions, approvals, events, progress adjustments, reward postings, AI recommendations and exports.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Board 11 - CMS, White-Label Content & SEO

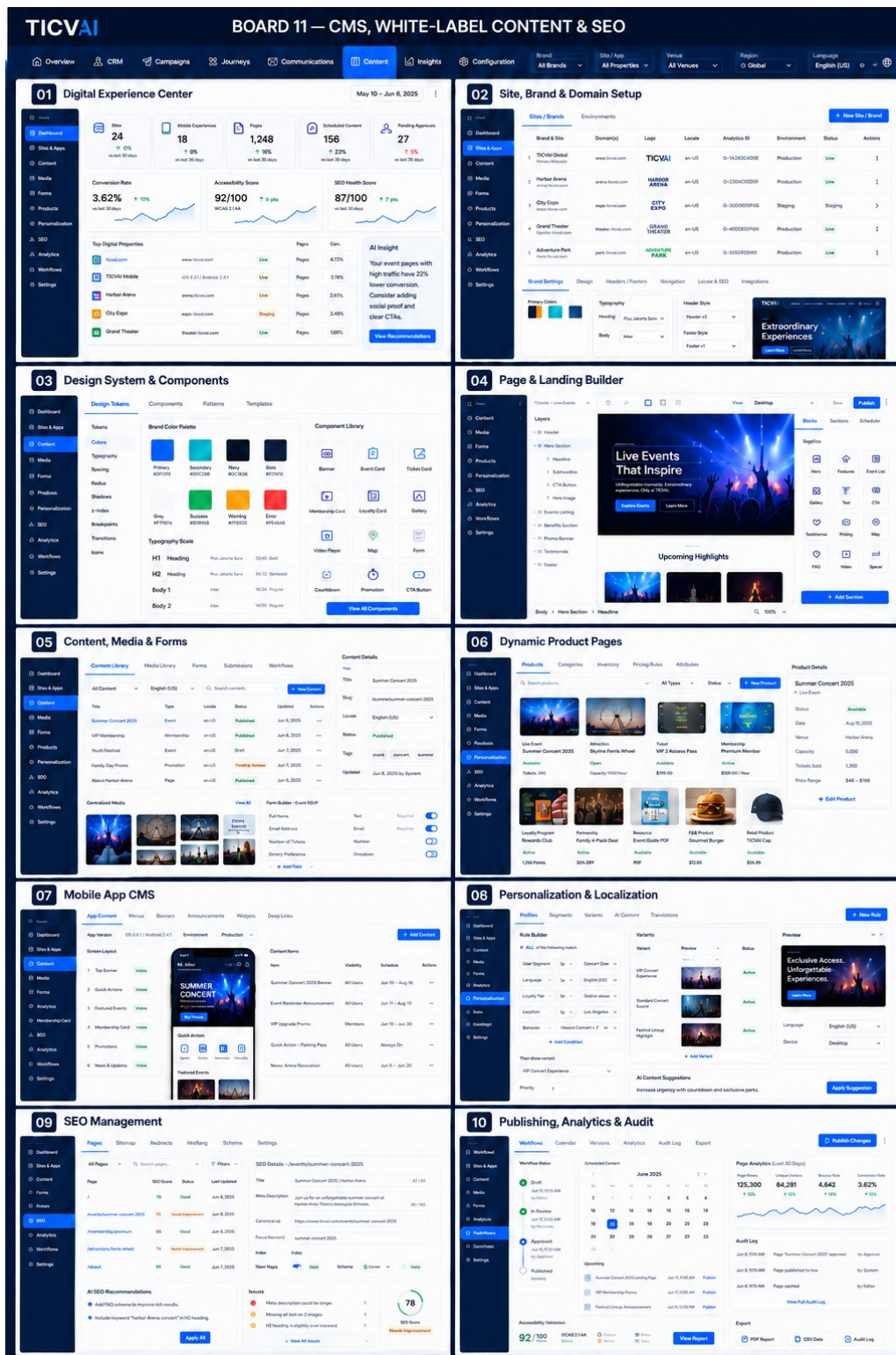


Figure 11. High-definition configuration board with all 10 screens.

Board 11 Scope of Work

Purpose. Provide a multi-site and mobile content platform for white-label digital experiences, live product content, personalization, localization, accessible publishing and SEO.

Source coverage. Matrix sections 22.10 - Marketing/CMS and 22.11 - SEO Management & Optimization

Operating workflow. Configure property and brand -> assemble design system -> create content/page/app experience -> bind products and personalization -> review accessibility/SEO -> approve and publish -> analyze -> version, roll back and audit.

Mandatory integrations. Ticketing, events, membership, loyalty, wallet, resources, retail, F&B, promotions, Customer Master, segmentation, mobile app, analytics, search engines, domains/CDN and media storage.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Digital Experience Center

Purpose: Monitor websites, portals, mobile experiences and content health.

- Show sites, apps, pages, scheduled content, approvals, conversion, accessibility score and SEO health.
- List digital properties with environment, brand, domain, locale, status, owner and performance.
- Surface broken links, stale content, publishing errors, low-performing pages and compliance issues.
- Provide explainable AI content and SEO opportunities with drill-down to affected pages.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Site, Brand & Domain Setup

Purpose: Configure multiple white-label digital properties from one deployment.

- Create websites, portals, microsites and app experiences by tenant, brand, venue, destination and business unit.
- Configure domains, environments, logos, colors, typography, icons, headers, footers and navigation.
- Set locales, default language, analytics IDs, consent/cookie settings, integrations and ownership.
- Validate domain, certificate, branding completeness and environment promotion before go-live.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Design System & Components

Purpose: Maintain reusable visual tokens and content components.

- Configure color, type, spacing, radius, shadow, breakpoint, transition and icon tokens by brand.

- Provide banner, event, ticket, membership, loyalty, gallery, video, map, form, countdown, promotion and CTA components.
- Define component variants, data properties, allowed placement, accessibility and responsive behavior.
- Version and approve changes and show every page/app experience affected before publication.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Page & Landing Builder

Purpose: Allow non-technical users to build pages and campaign landing experiences.

- Provide drag-and-drop sections, layer tree, reusable templates and desktop/tablet/mobile preview.
- Support content, product widgets, forms, countdowns, personalization, analytics tags and deep links.
- Configure slug, navigation, visibility, start/end schedule, campaign association and conversion goal.
- Validate responsive layout, required content, links, accessibility, SEO and approval before publish.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Content, Media & Forms

Purpose: Centrally manage reusable content, assets and data-capture forms.

- Maintain multilingual content entries, metadata, tags, owner, status, schedule and workflow.
- Manage images, video, PDFs, icons, audio and documents with rights, alt text, renditions and usage references.
- Build forms for registration, inquiry, waiver, survey, lead and custom workflows with validation and consent.
- Secure submissions, route them to approved services and apply retention, export and audit controls.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Dynamic Product Pages

Purpose: Publish live transactional content directly from TICVAI product services.

- Configure dynamic pages/widgets for events, attractions, tickets, capacity, pricing and availability.
- Display membership benefits/prices, loyalty tiers/rewards, wallet offers, resources, F&B and retail products.
- Define filters, sorting, related content, upsell/cross-sell, sold-out fallback and cache/refresh behavior.
- Ensure displayed price, inventory, eligibility and purchase actions come from authoritative shared engines.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Mobile App CMS

Purpose: Manage guest mobile-app content and navigation without releasing code.

- Configure home-screen layout, menus, banners, announcements, quick actions, widgets and featured content.
- Define deep links, audience visibility, language, schedule, app version and offline/cache behavior.
- Preview content for supported device sizes and validate inaccessible, missing or unsupported components.
- Publish through approval and staged rollout with version, rollback and analytics tracking.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Personalization & Localization

Purpose: Deliver relevant content variants under clear governance.

- Build visibility and content rules by segment, profile, language, loyalty, membership, geography and behavior.
- Configure variant priority, conflict resolution, fallback and real-guest/test-profile preview.
- Use AI for draft content and translation with glossary, brand tone, protected terms and manual review.
- Enforce consent and minimum-audience rules and audit the rule/version that produced each experience.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - SEO Management

Purpose: Configure technical and content SEO across dynamic and authored pages.

- Manage titles, descriptions, canonical URLs, Open Graph, keywords, slugs, aliases and 301/302 redirects.
- Generate XML sitemaps, hreflang and Schema.org markup for events, products, reviews, FAQs and organizations.
- Analyze readability, missing metadata, duplicates, broken links, crawl/index issues and internal-link opportunities.
- Provide AI recommendations with review and report ranking, organic traffic, conversion and revenue.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Publishing, Analytics & Audit

Purpose: Govern content release and measure digital-experience performance.

- Support draft, review, approval, scheduled publication, expiration, environment promotion and emergency unpublish.
- Maintain versions, visual/text comparison, rollback, accessibility validation and dependency checks.

- Report page/app views, engagement, conversion, product sales, search behavior and personalized-variant results.
- Audit authored and AI-generated content, translations, approvals, publications, rollbacks and administrative actions.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.*

Board 12 - Digital Waivers & Signature Management

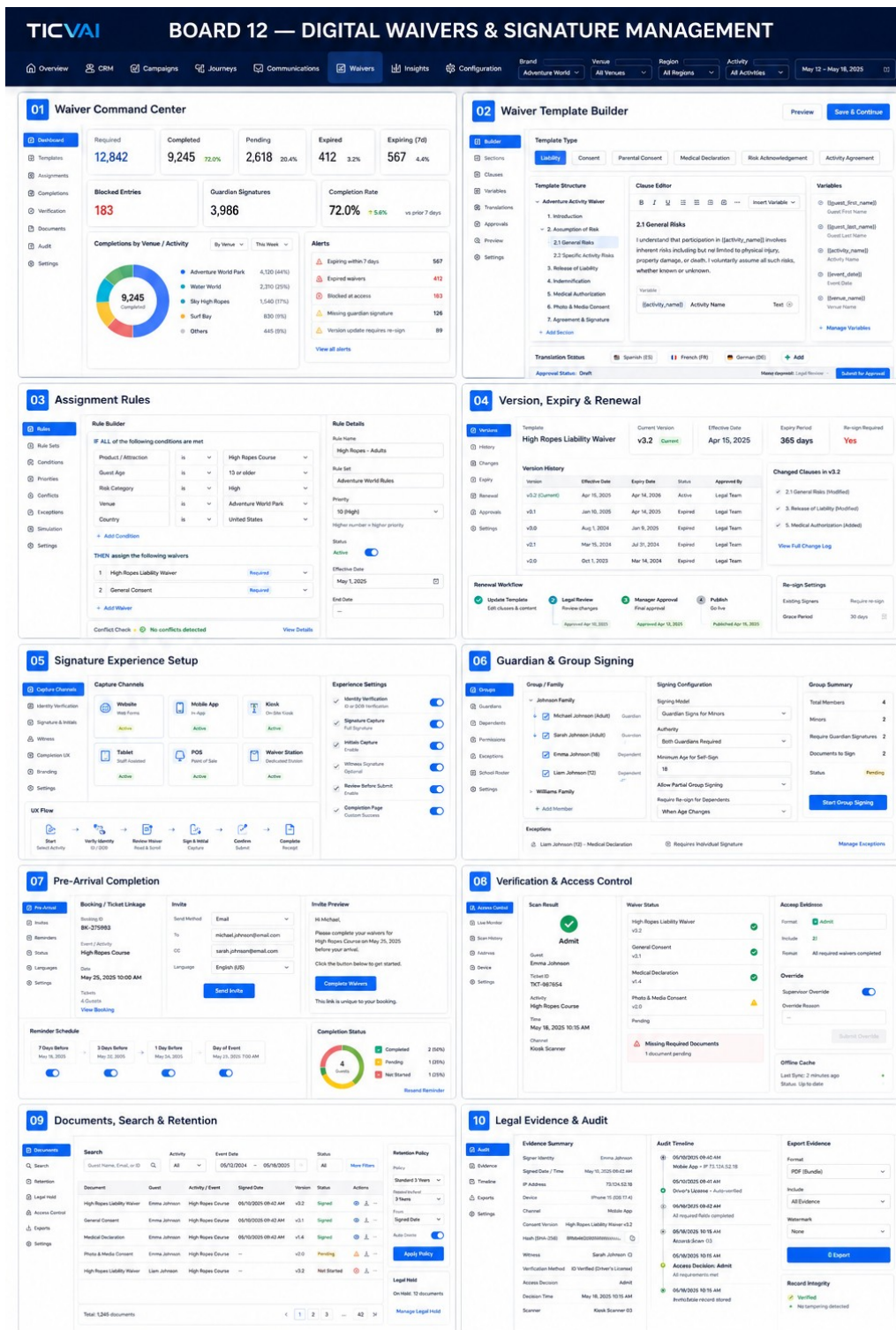


Figure 12. High-definition configuration board with all 10 screens.

Board 12 Scope of Work

Purpose. Create, assign, capture, verify and retain legally evidential digital waivers and consent forms across guest, family, group and access-control journeys.

Source coverage. Matrix section 22.12 - Digital Waivers & Signature Management

Operating workflow. Build and approve template -> assign by rules -> invite or present at channel -> verify signer/authority -> capture signature -> store evidence -> check at arrival/access -> renew when required -> retrieve and audit.

Mandatory integrations. Customer and family profiles, ticketing, reservations, events, memberships, website, mobile app, kiosk/tablet/POS, access control, notifications, document storage, retention/legal hold and audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Waiver Command Center

Purpose: Monitor waiver completion, expiry and operational access risk.

- Show required, completed, pending, expired and expiring waivers, blocked entries and guardian signatures.
- Analyze completion by venue, activity, event, product, channel, guest type and arrival date.
- Surface missing signatures, failed verification, version changes requiring re-sign and access-control exceptions.
- Provide operational drill-down and alerts without exposing document content to unauthorized roles.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 02 - Waiver Template Builder

Purpose: Create legal waiver and declaration templates without development.

- Support liability release, general consent, parental consent, medical declaration, risk acknowledgement and activity agreement types.
- Provide sections, clauses, variables, conditional content, initials, attachments and signature blocks.
- Manage multilingual content, legal owner, jurisdiction, risk category, approval and preview.
- Version every legal change and prohibit publishing until required legal and business approvals complete.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 03 - Assignment Rules

Purpose: Determine which waiver or form each guest must complete.

- Build rules using product, attraction, event, activity, venue, age, membership, risk category and jurisdiction.

- Assign one or multiple required documents with priority, effective dates and exception handling.
- Detect conflicting, missing or circular rules and simulate results for sample guests/bookings.
- Version, approve and audit rules and retain the rule/version used for each assignment.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 04 - Version, Expiry & Renewal

Purpose: Control legal-document lifecycle and re-sign requirements.

- Maintain version history, effective dates, changed clauses, approver, status and expiry period.
- Configure whether a change or elapsed period requires all, selected or future guests to sign again.
- Define renewal reminders, grace period, replacement behavior and treatment of active bookings.
- Preserve every signed version and prevent current templates from altering historical evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 05 - Signature Experience Setup

Purpose: Configure secure signing across supported guest touchpoints.

- Support website, mobile app, kiosk, tablet, POS and dedicated waiver-station channels.
- Configure identity verification, signature, initials, witness, review-before-submit and completion receipt.
- Define accessible, multilingual flow, timeout, session recovery, device controls and offline behavior.
- Test the end-to-end experience and map every captured element to the evidence record.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 06 - Guardian & Group Signing

Purpose: Manage authority and multi-participant signing for minors and groups.

- Use Customer Master family/guardian relationships and record the authority basis for each dependant.
- Support one authorized signer for multiple family members, students or group participants where permitted.
- Configure minimum self-sign age, one/both guardian requirement, partial group signing and exceptions.
- Show roster status and require individual signatures for documents or participants that cannot be delegated.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 07 - Pre-Arrival Completion

Purpose: Collect required waivers before the guest reaches the venue.

- Link the request to booking, ticket, event, activity, participants and scheduled arrival.

- Send secure email, SMS or WhatsApp link/QR in the guest's language with a completion deadline.
- Configure reminder schedule, resend, authentication, expiry and status synchronization.
- Show completed, pending and not-started participants and stop reminders immediately after valid completion.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 08 - Verification & Access Control

Purpose: Enforce waiver requirements during check-in and access validation.

- Return real-time valid, missing, expired or wrong-version status during ticket scan or check-in.
- Show required documents and participant while minimizing sensitive content at the gate.
- Configure admit/block outcomes, supervisor override, reason, approval and offline cache/freshness.
- Record scan, decision, waiver evidence reference, device, operator, override and final access outcome.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 09 - Documents, Search & Retention

Purpose: Securely store and retrieve signed waiver evidence.

- Search by guest, guardian, participant, booking, ticket, activity, event, date, version and status.
- Store signed PDF/evidence package, metadata, relationships, status and authorized view/download actions.
- Apply encryption, RBAC/PBAC, retention, legal hold, archive and deletion/anonymization policies.
- Track every access, download, policy action and downstream reference and prevent silent alteration.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Screen 10 - Legal Evidence & Audit

Purpose: Provide a defensible record of signing and subsequent operational use.

- Record signer and participant identity, authority, timestamp, timezone, IP/device, channel and verification method.
- Store consent/template version, content hash, signature/initials, witness, attachments and completion receipt.
- Include access-control decisions, overrides, renewals, revocations and related communications in the timeline.
- Export a controlled evidence bundle with integrity validation and full audit history.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the audit history.

Implementation Handover Checklist

Check	Area	Required Evidence
☐	Functional	All 120 screens are mapped to user stories, acceptance criteria and traceable matrix requirements.
☐	Architecture	Shared engines and services are reused; no duplicate customer, consent, promotion, loyalty, wallet, notification or audit logic is introduced.
☐	Data	Entities, identifiers, retention, migration, master-data ownership and reconciliation are documented.
☐	Security	RBAC/PBAC, SSO/MFA, field masking, SoD, temporary access, override and audit requirements are tested.
☐	Integrations	APIs, events, webhooks, provider connectors, retries, idempotency and monitoring are documented and validated.
☐	AI	Models/use cases, data access, explainability, confidence, human review, feedback, monitoring and governance are complete.
☐	UX & accessibility	Approved TICVAI design system, responsive behavior, keyboard support and WCAG 2.2 AA-aligned checks are complete.
☐	Quality	Unit, integration, end-to-end, performance, security, accessibility, localization and UAT evidence is available.
☐	Operations	Dashboards, alerts, logs, runbooks, support ownership, backup/recovery and incident procedures are ready.
☐	Handover	Source code, configuration, documentation, test assets, deployment instructions and knowledge transfer are complete.

END OF DOCUMENT